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Message



Shri Ashok Gehlot,
Hon'ble Chief Minister, Rajasthan

Ground water is an inseparable aspect in the lives of the people of Rajasthan. There are no major or perennial rivers flowing through the State. Moreover, most part of the state receives very less rainfall. Our dependence on ground water is extremely critical not only as drinking water for human beings and cattle but also as major source of irrigation.

The Government of Rajasthan is keen on sustainable utilization of the ground water. The State water policy, adopted in 2010 emphasizes on community participation in ground water management and use of information technology for widespread dissemination of data for the benefit of all sections of the society.

I am pleased to note that the Ground Water Department of the State has carried a detailed assessment of quality, quantity and distribution of ground water resources of the State through mapping of aquifers and using most modern tools and techniques.

It is also heartening to note that the outcome of the project is being published in the form of '*Hydrogeological Atlas of Rajasthan*'. It will incorporate the scientific findings and reliable information in various types of readily understandable GIS maps based on ground water potential viz. zone maps, depth to water level maps, ground water categorization maps, chemical quality maps, aquifer maps etc. These would be certainly found useful by experts and user agencies. The same would also help in raising the awareness levels of the various stakeholders in utilizing the precious ground water resources of the State.

I take this opportunity to congratulate the Ground Water Department officials for bringing this useful publication and wish them success in all such endeavor.



(Ashok Gehlot)

Message



Dr. Jitendra Singh,
Hon'ble Minister for Energy,
Non-Conventional Energy Resources,
PHED, GWD, Information & Public Relations

Ground water is a renewable resource but its long term sustainability depends on the balance of recharge and withdrawal ground water is added every year through rainfall and other sources. In Rajasthan, volume of water extracted for different purposes far exceeds its replenishment which has led to depletion of ground water in large parts of state.

While the surface water is visible and largely measurable, the ground water is hidden underneath. To meet the ever increasing demand of water, the state has been promoting conjunctive use of water. In order to have a state level estimation of water resources, the estimation of ground water quality and quantity is a key input. The studies being very scientific in nature it remains a challenge how to communicate the message to all stakeholders.

The publication of '*Hydrogeological Atlas of Rajasthan*' is a major step forward and will help in visualization of ground water related information in very simplified graphic and tabular forms. The effort that has gone into creation of the atlas from basic information in archives is very much appreciated.



15.4.13.

(Jitendra Singh)

Message



Dr. Purushottam Agarwal

Principal Secretary to Government,
Public Health Engineering and Ground Water Department, Rajasthan

Ground water has been the principal source over years to meet the needs of human and cattle population of the state. Increasingly modern techniques are used for extraction of large volume of ground water which has exceeded recharge in most parts of the State. As a result, water table in most of the areas has gone down significantly. People are digging deeper and deeper to reach the ground water to meet their increasing needs. Maintaining a healthy practice of balance between recharge and withdrawal is critical for its sustainability.

The state had initiated various programs under which a systematic assessment of water resources is being carried out and all future water management plans are chalked out on the basis of sound database and historic data analysis. Increasing population, industrialization, agricultural need, power generation etc. are bound to pose challenges before us in coming years and to face them we have documented 'Water Resource Vision 2045' which highlights the short term (upto 2015) and long term (upto 2045) thrust areas and action plan which are pre-requisites for successful implementation of the State Water Policy. We are committed to achieving the objective of optimum use of every drop of scarce and precious utilizable water resource. In this regard, the Ground Water Department has also carried out a systematic assessment of ground water resources and three dimensional mapping of aquifers that gives a new insight.

I congratulate the Ground Water Department on publication of '*Hydrogeological Atlas of Rajasthan*' which emphasizes on various aspects of ground water which I feel will help us in spreading the information on current scenario and consequently help in managing it better.

(Purushottam Agarwal)

Foreword



Arno Schaefer

Minister Counselor, Head of Operations

The Government of Rajasthan has entered into an agreement with the European Union through the State Partnership Programme for implementing state-wide water sector reform leading to sustainable integrated water resources management in Rajasthan, and supporting the Panchayat Raj institutions in executing their responsibilities in equitable access to safe, adequate, affordable and sustainable water supply as well as conservation, stabilization and replenishment of surface and ground water.

Ground water is the primary source for drinking, domestic, agriculture and industries and other purposes in Rajasthan. The stage of ground water development is close to 135% i.e., if the net annual ground availability in Rajasthan is about 10.7 billion cubic meter, the annual draft is 14.5 billion cubic meter. As a result of this, the water levels have been depleting alarmingly. The ground water in 166 out of 249 blocks of the state fall under the category of over-exploited. This indiscriminate and sometime exploitative use of ground water has led to questions regarding its sustainability.

This publication '*Hydrogeological Atlas of Rajasthan*' is the culmination of a long term effort of collation, integration, interpretation and analysis of vast archive of historic data obtained by Rajasthan Ground Water Department. The atlas is in three parts, (1) Statewide Atlas, which shows the state of ground water, water quality, current stage of exploitation, aquifer mapping and basic information on administrative setup, climate, geology and geomorphology; (2) A detailed series, wherein compilation of information and district level thematic maps and all 33 districts have been separately presented and (3) Compilation of information at the River Basin level. In addition to the thematic maps, a large number of cross sections and the 3 dimensional features of aquifers have also been presented.

I am convinced that the Atlas will contribute to better understand ground water and help managerial and future planning. The current Geographic Information System (GIS) based study is a pioneering step in India and will pave the way for similar approach to be adopted by other Indian States.

I appreciate very much the efforts made by the Ground Water Department, Government of Rajasthan, The European Union Technical Assistance Team and M/s Rolta India Limited for undertaking this study and compiling this Atlas under the India-EU State partnership Programme.

(Arno Schaefer)

Hydrogeological Atlas of Rajasthan



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2013

ADMINISTRATIVE SETUP

Location and Administrative Set-up:

Rajasthan is situated in the Northwestern part of India and shares its border with Punjab, Haryana, Uttar Pradesh, Madhya Pradesh, and Gujarat among Indian states. The western border is shared with Pakistan. The state capital is Jaipur city and is administratively divided into 33 districts and 249 blocks. As per 2011 census, Rajasthan has total population of is 6,84,59,588. It covers approximately 3,42,239 square kilometers of area and extends between 22° 58' 09.75" to 30° 12' 05.27" North latitude and 69° 18' 41.91" to 78° 22' 31.11" East longitudes.

District Name	Ajmer	Alwar	Banswara	Baran	Barmer	Bharatpur	Bhilwara	Bikaner	Bundi	Chittaurgarh	Churu	Dausa	Dhaulpur	Dungarpur	Ganganagar	Hanumangarh	Jaipur	Jaisalmer
Area(sq km)	8,484.08	8,382.86	4,493.05	6,993.94	28,550.95	5,079.47	10,445.18	30279.26	5,776.48	7,837.04	13,844.05	3,417.65	3,040.33	3,770.78	10,683.97	9,929.25	11,136.02	38,487.17
Number of Blocks	8	14	8	7	8	9	11	6	5	11	6	5	4	5	8	7	13	3
Number of Towns	9	9	3	6	2	9	8	4	7	6	10	5	3	3	12	6	11	2
Number of Villages	1,038	1,994	1,410	1,207	1,941	1,472	1,745	875	849	1,707	882	1,058	802	858	3,014	1,905	2,131	637
Population (Per 2011 Census)	2,584,913	3,671,999	1,798,194	1,223,921	2,604,453	2,549,121	2,410,459	2,367,745	1,113,725	1,544,392	2,041,172	1,637,226	1,207,293	1,388,906	1,969,520	1,779,650	6,663,971	672,008

District Name	Jalor	Jhalawar	Jhunjhunun	Jodhpur	Karauli	Kota	Nagaur	Pali	Pratapgarh	Rajsamand	Sawai Madhopur	Sikar	Sirohi	Tonk	Udaipur	Total
Area (sq km)	10,647.40	6,315.27	5,911.03	22,696.90	4,985.11	5,122.26	17,805.10	12,378.91	4,400.66	4,629.38	5,051.96	7,726.75	5,139.03	7,190.38	11,772.83	3,42,405.50
Number of Blocks	8	6	8	10	5	5	11	10	5	7	5	8	5	6	12	249
Number of Towns	3	8	13	4	3	11	12	11	3	5	4	9	5	7	9	222
Number of Villages	706	1,600	859	1,063	798	892	1,500	949	943	987	794	992	462	1,093	2,190	41,353
Population (Per 2011 Census)	1,830,151	1,411,327	2,139,658	3,685,681	1,458,459	1,950,491	3,309,234	2,038,533	706,807	1,158,283	1,338,114	2,677,737	1,037,185	1,421,711	3,067,549	5,45,36,327

Climate:

The climate of Rajasthan can be divided into four seasons: Pre-Monsoons, Monsoon, Post-Monsoon and Winter. Pre-monsoon, which extends from April to June, is the hottest season, with temperatures ranging from 32°C to 45°C. In western Rajasthan the temperature may rise to 48°C, particularly in May and June. In the desert regions, the temperatures drop in night significantly. Winds generally flow from the west and sometimes carry dust storms.

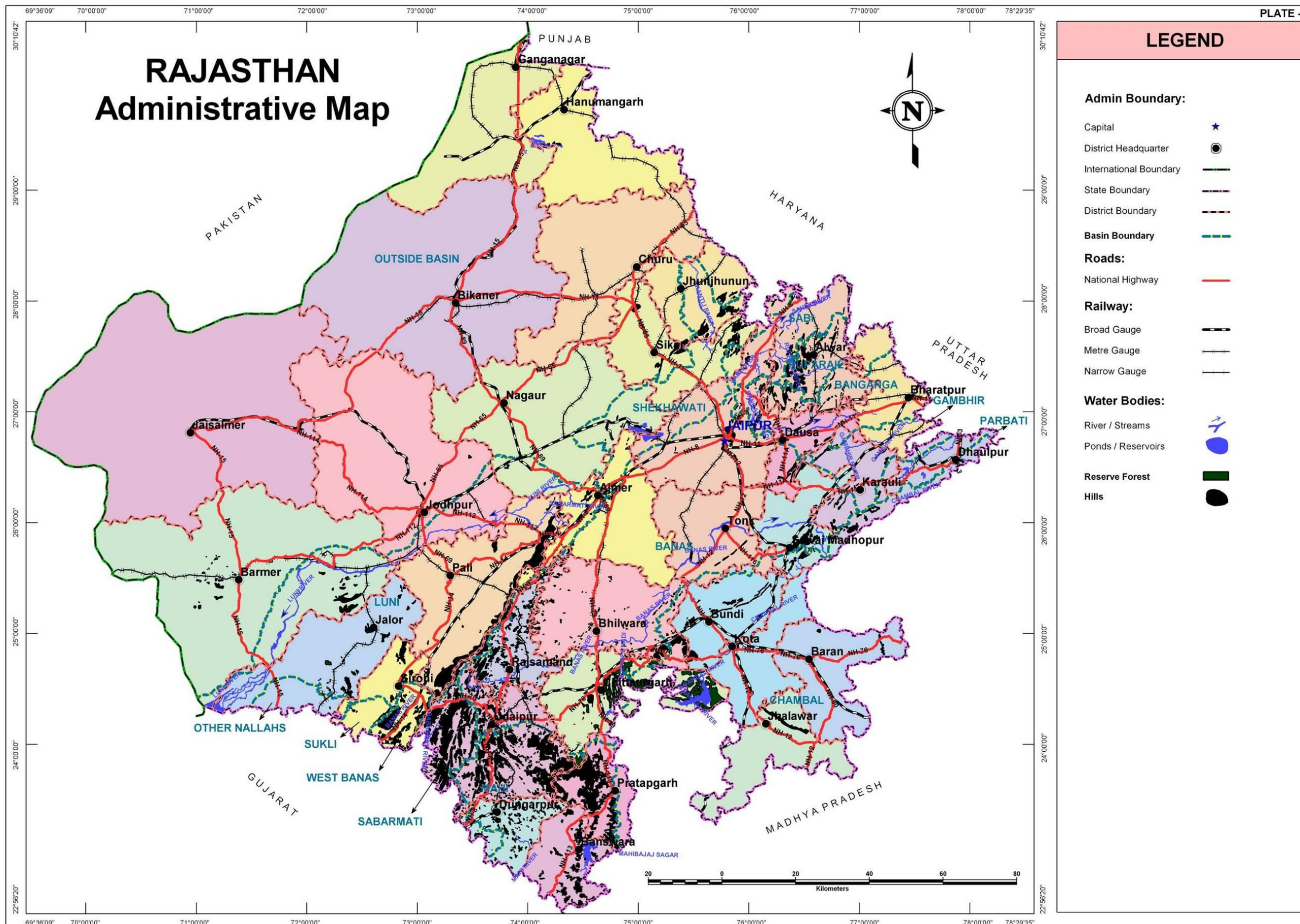
The second season Monsoon extends from July to September, temperature drops (35°C to 40°C) but humidity shows rise. Most of the rainfall occurs during this period. The Post-monsoon period is from Oct to December. The average maximum temperature is 33°C to 38°C, and the minimum is between 18°C and 20°C.

The fourth season is the winter, from January to March. There is a marked variation in maximum and minimum temperature and regional variations across the state. January is the coolest month of the year. And temperature may drop to 0°C in some cities of Rajasthan, like Churu and Mt. Abu. There is slight precipitation in the north and north-eastern region of the state, and light winds, predominantly from the north and north-east.

The Eastern Rajasthan has sub-humid to semi-arid climate whereas that in the west is even arid marked by Great Thar Desert. Total average annual rainfall in Rajasthan in 2010 was around 685 mm.



PLATE - I



TOPOGRAPHY

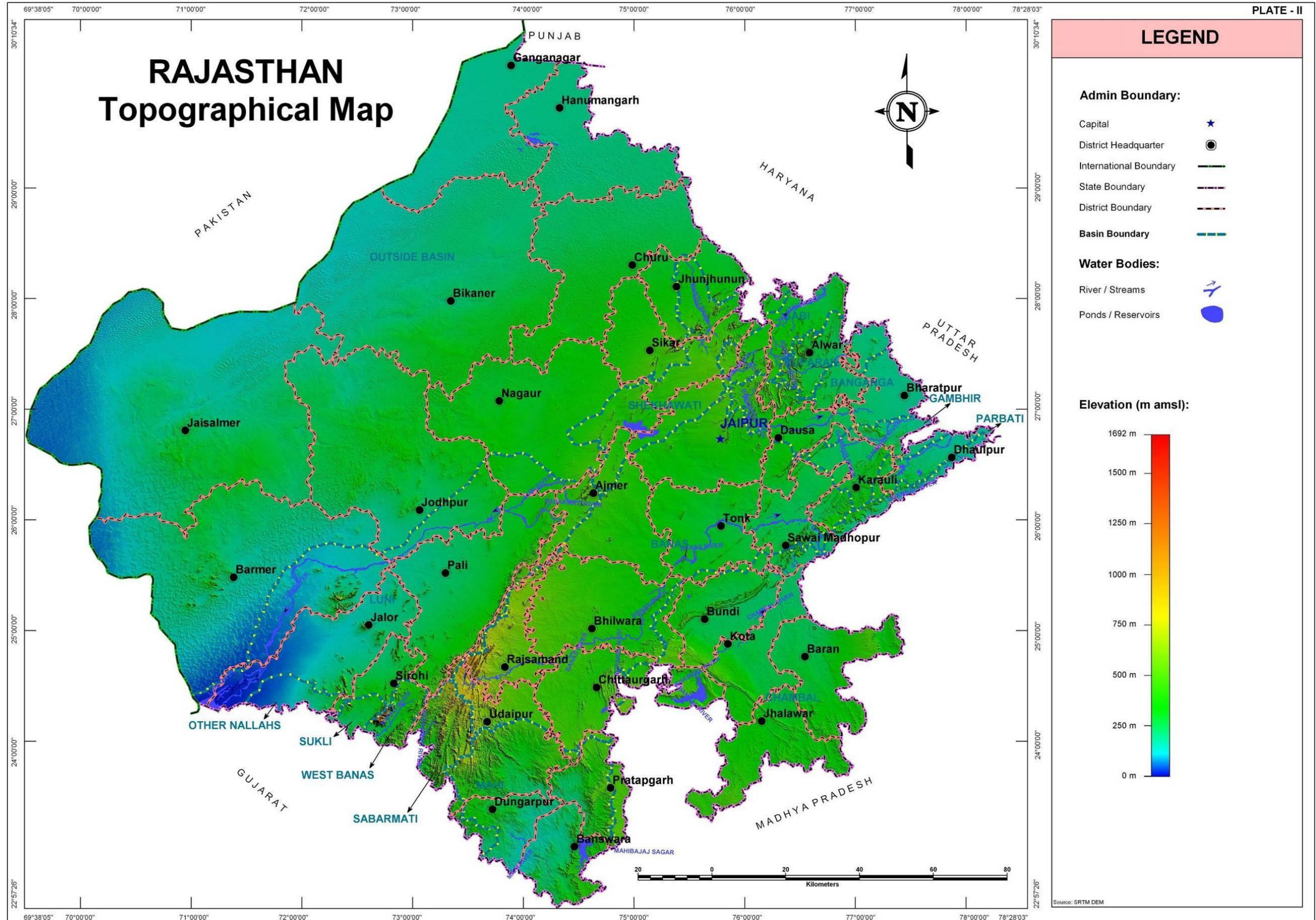
The state has been witness to long geological past and the same is reflected in its physiography. There is a NE-SW trending ridgeline, the Aravalli range exposing some of the oldest rocks in the country forming hills, whereas the western part is Quaternary wind-blown sand forming deserts. In the Northeast, the vast plains are formed by large east flowing rivers and their sediments. The plateaus in the Southeast are formed in Vindhyan sedimentary rocks and Deccan Basalts.

Most of the areas in the western part occupied by Thar Desert do not have any well-defined drainage system while the south and eastern parts have developed good river systems drained by the Luni, Banas, Chambal, Banganga etc. Aravallis form major drainage divide with majority of rivers originating from it flowing towards east and very few flowing towards west.

Hills passing through Udaipur, Sirohi, Sikar, Sawai Madhopur, Rajsamand, Pali and Jhunjhunun reach very high elevations of beyond 1000m above mean sea level (amsl). Low hills are also seen to be formed in Vindhyan in the Southeastern part of the area that reach an elevation of 500-600m amsl. Scattered hills are also seen in Jalor, Barmer and Jodhpur.

The plains are formed in the general elevation range of 200-300m amsl in Bharatpur, Dausa, Jaipur, Tonk, Ajmer and Bhilwara region. Western parts of Bikaner and Jaisalmer form dunal sandy plains which are in general around 100m amsl and below, occasionally reaching even 0m amsl.

S. No	District Name	Min Elevation (m amsl)	Max Elevation (m amsl)
1	Ajmer	301	870
2	Alwar	190	771
3	Banswara	113	581
4	Baran	200	549
5	Barmer	0	932
6	Bharatpur	157	416
7	Bhilwara	305	822
8	Bikaner	104	349
9	Bundi	200	547
10	Chittaurgarh	300	616
11	Churu	200	471
12	Dausa	203	596
13	Dhaulpur	111	341
14	Dungarpur	112	546
15	Ganganagar	105	230
16	Hanumangarh	164	239
17	Jaipur	271	785
18	Jaisalmer	37	347
19	Jalor	0	967
20	Jhalawar	250	524
21	Jhunjhunun	260	1,035
22	Jodhpur	150	448
23	Karauli	131	523
24	Kota	176	518
25	Nagaur	275	794
26	Pali	149	1,068
27	Pratapgarh	162	577
28	Rajsamand	376	1,294
29	Sawai Madhopur	164	541
30	Sikar	294	1,032
31	Sirohi	184	1,689
32	Tonk	216	600
33	Udaipur	160	1,290



RAINFALL

The rainfall is very scanty and erratic. The individual rain gauge station data of year 2010 shows very wide variation from less than 100mm to more than 1000mm of total annual rainfall in the state, whereas district level averages reveal that the minimum to be about 360mm in Ganganagar to a maximum of 948mm in Sirohi district. To the east of Aravalli Range, the annual total rainfall is, in general, higher and averages above 600mm whereas the districts in the west of Aravalli receive very less rainfall, i.e., in the range of less than 300mm to 400mm annually minimum being in Ganganagar and Jaisalmer districts. The trend of distribution is reflected in isohyet maps presented in Plate III,

District Name	Total Monsoon Rainfall (mm)	Total Non-Monsoon Rainfall (mm)	Total Annual Rainfall (mm)*
Ajmer	532	124	656
Alwar	732	51	783
Banswara	651	35	687
Baran	562	51	613
Barmer	475	48	524
Bharatpur	729	70	799
Bhilwara	610	156	767
Bikaner	573	23	596
Bundi	583	105	689
Chittaurgarh	729	99	828
Churu	621	46	667
Dausa	639	101	739
Dholpur	670	94	763
Dungarpur	605	61	665
Ganganagar	337	23	360
Hanumangarh	527	35	562
Jaipur	735	95	830
Jaisalmer	357	16	373
Jalore	658	125	783
Jhalawar	587	54	641
Jhunjhunu	705	63	768
Jodhpur	395	38	434
Karauli	628	114	742
Kota	563	55	618
Nagaur	531	40	571
Pali	552	123	675
Pratapgarh	707	40	747
Rajsamand	653	187	840
Sawai Madhopur	668	142	810
Sikar	763	74	836
Sirohi	784	164	948
Tonk	630	130	760
Udaipur	780	102	882
Statewide average	614	816	695

*District level total rainfall is computed from 2010-2011 Monsoon season data of raingauge stations averaged at district level. (Source of data: <http://waterresources.rajasthan.gov.in>)



LEGEND

Admin Boundary:

- Capital ★
- District Headquarter ●
- Raingauge Stations ●
- International Boundary —
- State Boundary —
- District Boundary —
- Basin Boundary —

Roads:

- National Highway —

Railway:

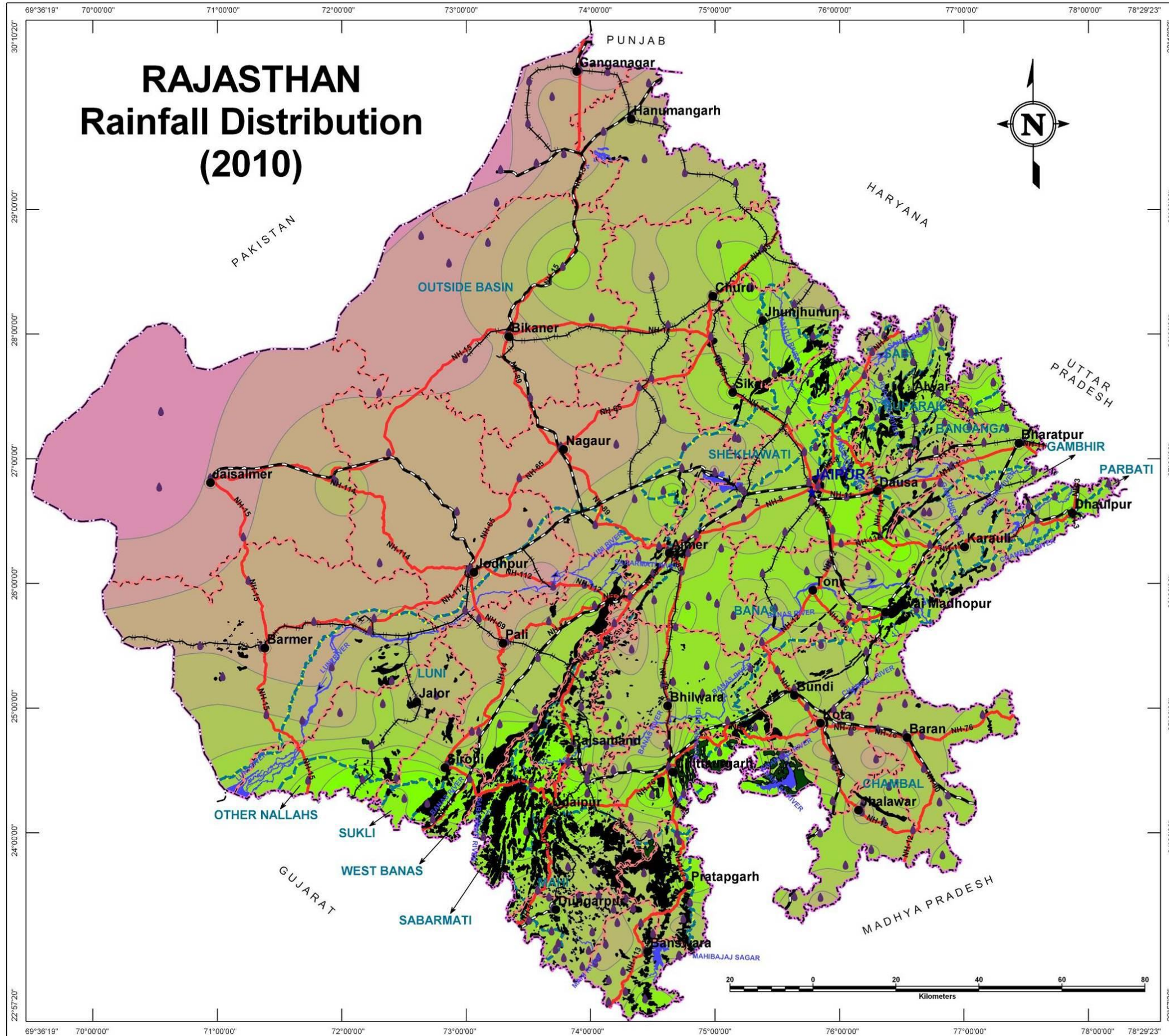
- Broad Gauge —
- Metre Gauge —
- Narrow Gauge —

Water Bodies:

- River / Streams —
- Ponds / Reservoirs ●
- Reserve Forest —
- Hills —

Isohyets (mm):

- 200-300
- 300-400
- 400-500
- 500-600
- 600-700
- 700-800
- 800-900
- 900-1000
- > 1000

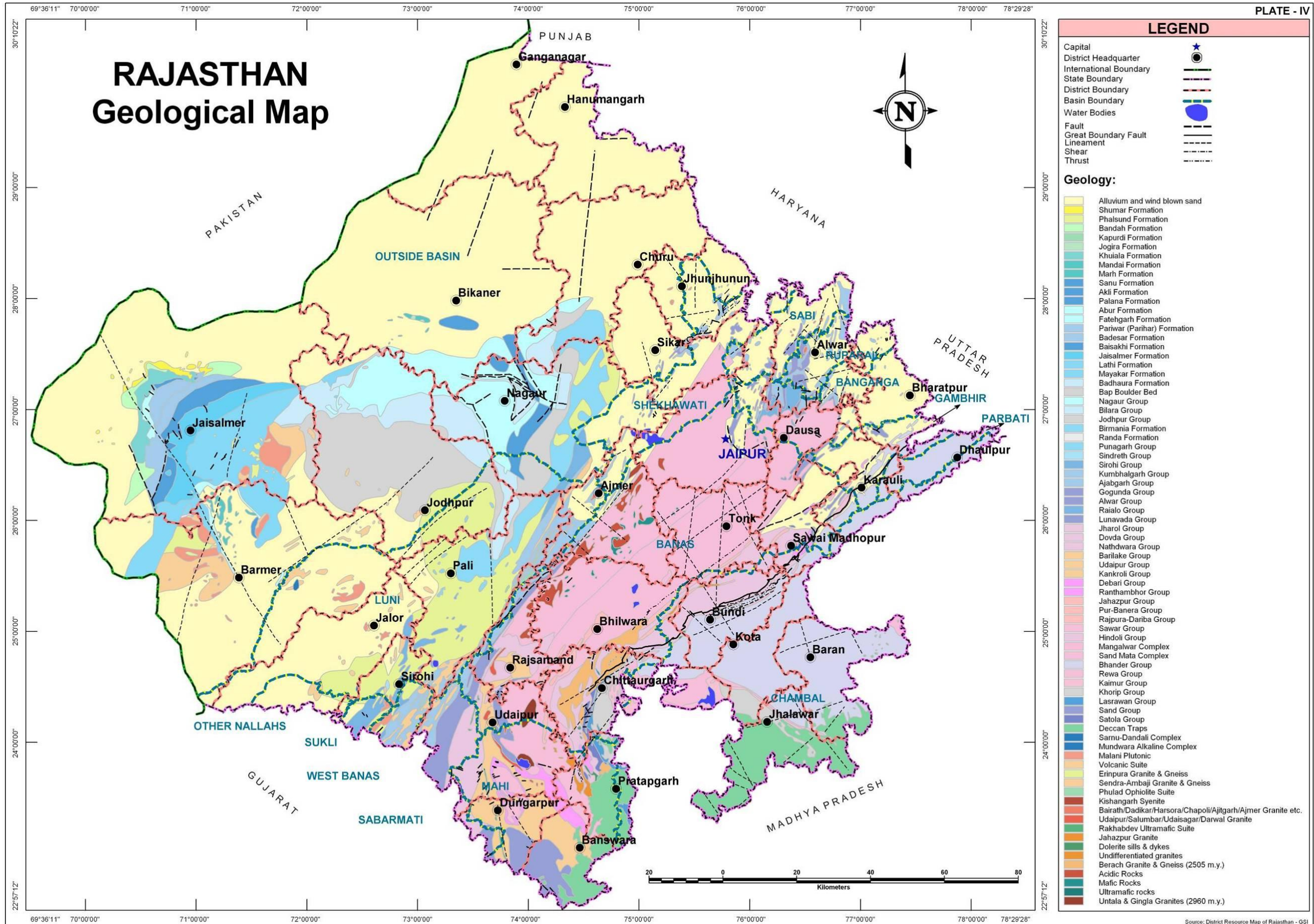


Rajasthan is endowed with a Geological sequence of rocks from the oldest Archaean, Metamorphites, represented by Bhilwara Super Group (more than 2500my old) to sub-recent, alluvium and windblown sand. The western and north-western parts of the state are covered by vast blanket of young unconsolidated deposits including the blown sand of the Thar Desert of western Rajasthan. The remaining area exposes wide variety of hard rocks, which include various types of metamorphic schists, quartzites, marbles and gneisses of Pre-Cambrian age with associated acid, and basic intrusive rocks. The semi-consolidated sedimentary rocks include those of Aravali Super group, Delhi Super group, Upper Precambrian, Vindhyan Super group and Cambrian to Jurassic, Cretaceous and Tertiary age. The southeastern extremity of the state is occupied by a pile of basaltic flows of Deccan Traps of Cretaceous age. Several mineral deposits of economic importance occur in association with the above rock units.

The Aravalis, one of the most ancient mountains in the world, have the oldest granitic and gneissic rocks at their base, overlain by the rocks of the Aravali Super group, Delhi Super group, the Vindhyan Super group and younger rocks. These rocks are highly metamorphosed at certain places. The Aravali Super group is a vast formation composed of basalt, quartzites, shales, conglomerates, composite gneisses and slates. The Delhi Super group is divided into lower Raialo group, middle Alwar group and Upper Ajabgarh group. Raialo group is rich in crystalline limestones, grits, schistose rocks and quartzites. Alwar group and Ajabgarh group consist mostly of calc-silicates, quartzites, grits and schistose rocks. The other important lithological formations consist of a thick series of sedimentary rocks comprising sandstone, limestone and shale. These have been classified as upper and lower Vindhyan. The deposition of these rocks in western Rajasthan was preceded by igneous activity, which included a thick pile of lava, mostly of an acidic nature. The plutonic equivalent of these lavas is seen in the form of granite bosses and sills in Jalore, Siwana, Mokalsar and Jodhpur areas. Rocks of the above mentioned igneous activity have been designated as Erinpura granite and Malani igneous suite.

Table: Generalized geologic succession (after Heron, 1953) and lithology of Rajasthan

Recent & Sub recent		Alluvium and blown sand
Vindhyan System		Sandstone, limestone & boulders
	Malani Rhyolites	Granite, Ultrabasic rock, Erinpura granite, permatite, aplite Epidiorite and hornblende schist
Delhi System	Ajabgarh Series	Upper phyllite, Limestone, Biotitic limestone and calc-schist, Phyllites, biotite schist and composite gneiss
	Alwar Series	Quartzites, Arkose, grit and conglomerates
	Raialo Series	Garnetiferous biotite schist, Limestone (marble), Local basal grit
Aravali System		Impure limestone, quartzites, phyllites, biotite-schist, composite gneiss, Quartzites, grits and local soda-syenites, conglomerate, Local amygdaloids and tuffs
	Banded Gneissic Complex (BGC)	Schists, gneisses and composite gneiss, Quartzites



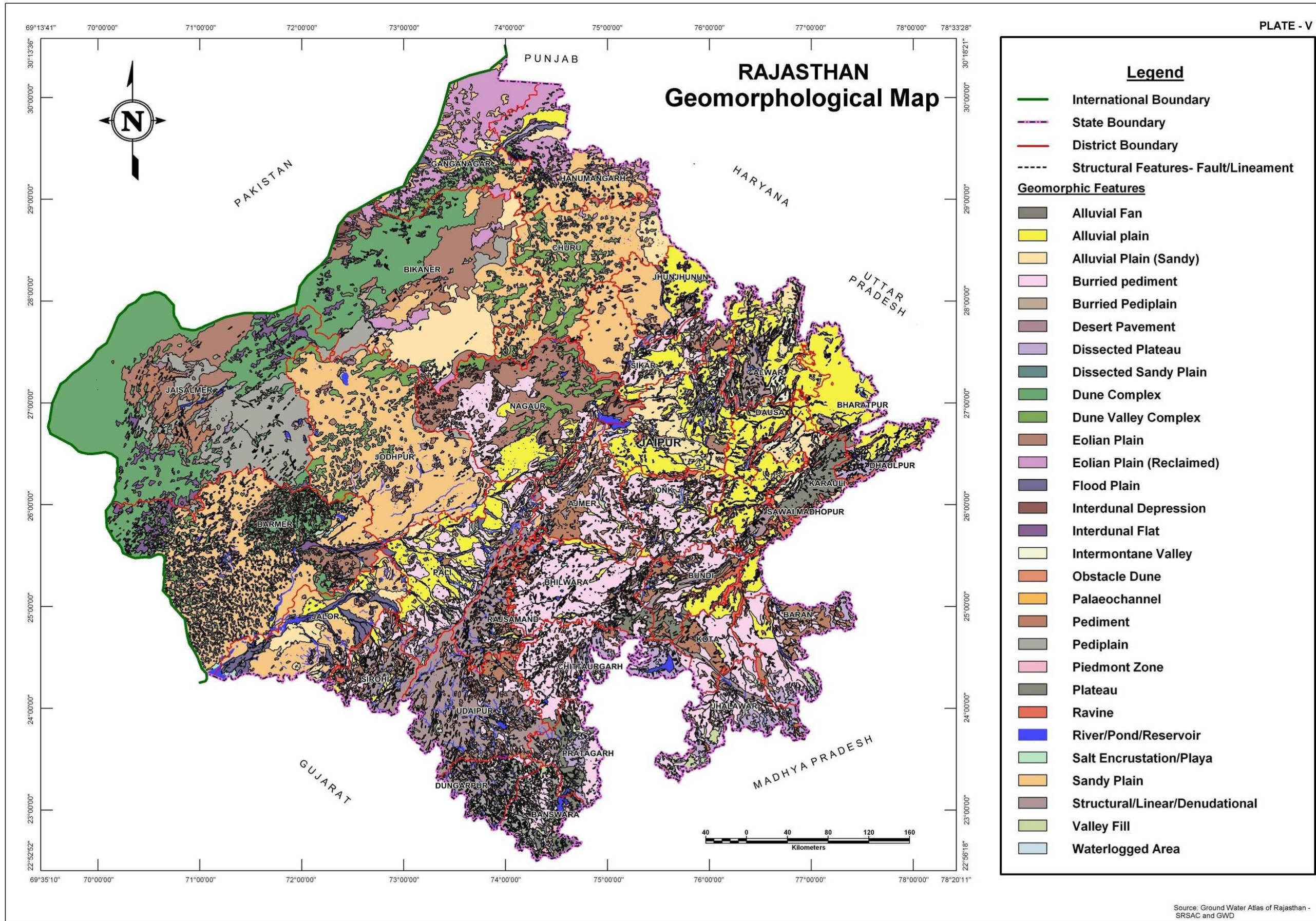
GEOMORPHOLOGY

The geomorphology of the state is quite varied. Broadly three types of units occupy major part of the state, viz, Aeolian, Fluvial and to some extent denudational. Aravallis form central NE-SW trending axis occurring as structural ridges and linear ridges in the state and the eastern/southeastern part represented by Vindhyaans.

Origin	Landform Unit	Description
Aeolian	Desert Pavement	Flat barren stony desert plain having coarse, angular, wind polished gravel and small stones lying on rock cut surface.
	Dissected Plateau	Plateau, criss-crossed by fractures forming deep valleys.
	Dissected Sandy Plain	Sandy plain highly dissected by stream and drainage
	Dune Complex	An undulating plain composed of number of sand dunes of cressent shape.
	Dune Valley Complex	Cluster of dunes and interdunal spaces with undulating topography formed due to wind-blown activity, comprising of unconsolidated sand and silt.
	Eolian Plain	Formed by aeolian activity, with sand dunes of varying height, size, slope. Long stretches of sand sheet. Gently sloping flat to undulating plain, comprised of fine to medium grained sand and silt. Also scattered xerophytic vegetation.
	Eolian Plain (Reclaimed)	Gently sloping with sheet of sand or sand dunes, scattered xerophytic vegetation.
	Interdunal Depression	Slightly depressed area in between the dunal complex showing moisture and fine sediments.
	Interdunal Flat	Flat, narrow land between dunes.
	Obstacle Dune	Formed on windward/leeward sides of obstacle like isolated hills or continuous chain of hills, dune to obstruction in path of sand laden winds. Badly dissected well cemented and vegetated.
	Sandy Plain	Formed of aeolian activity, wind-blown sand with gentle sloping to undulating plain, comprising of coarse sand, fine sand, silt and clay.
Denudational	Burried Pediment	Pediment covers essentially with relatively thicker alluvial, colluvial or weathered materials.
	Intermontane Valley	Depression between mountains, generally broad & linear, filled with colluvial deposits.
	Pediment	Broad gently sloping rock flooring, erosional surface of low relief between hill and plain, comprised of varied lithology, criss-crossed by fractures and faults.
	Pediplain	Coalescence and extensive occurrence of pediment.
	Pediment Zone	Formed by coalescence of several alluvial fans by stream covering large area at foot hills, with gentle slope in humid to sub humid region.
Fluvial	Alluvial Fan	A fan shaped mass of sediment deposit at a point along a Nallah, river where there is a decrease in gradient.
	Alluvial Plain	Mainly undulating landscape formed due to fluvial activity, comprising of gravels, sand, silt and clay. Terrain mainly undulating, produced by extensive deposition of alluvium.
	Alluvial Plain (Sandy)	Flat to gentle undulating plain formed due to fluvial activity, mainly consists of gravels, sand, silt and clay with unconsolidated material of varying lithology, predominantly sand along river.
	Flood Plain	The surface or strip of relatively smooth land adjacent to a river channel formed by river and covered with water when river over flows its bank. Normally subject to periodic flooding.
	Paleochannel	Mainly buried on abandoned stream/river courses, comprising of coarse textured material of variable sizes.
	Valley Fill	Formed by fluvial activity, usually at lower topographic locations, comprising of boulders, cobbles, pebbles, gravels, sand, silt and clay. The unit has consolidated sediment deposits.
	Ravine	Small, narrow, deep, depression, smaller than gorges, larger than gulley, usually carved by running water.
	Salt Encrustation/Playa	Topographical depression comprising of clay, silt, sand and soluble salts, usually undrained and devoid of vegetation.
	Water logged/ Wetland	Area submerged in water or area having very shallow water table. So that it submerges in water during rainy season.
Hills	Denudational Hill	Steep sided, relict hills undergone denudation, comprising of varying lithology with joints, fractures and lineaments.
	Linear Ridge	Long narrow low-lying ridge usually barren, having high run off may form over varying lithology with controlled strike
	Plateau	Formed over varying lithology with extensive, flat, landscapes, bordered by escarpment on all sides. Essentially formed horizontally layered rocky marked by extensive flat top and steep slopes. It may be criss crossed by lineament.
	Structural Hill	Linear to arcuate hills showing definite trend-lines with varying lithology associated with folding, faulting etc.



PLATE - V

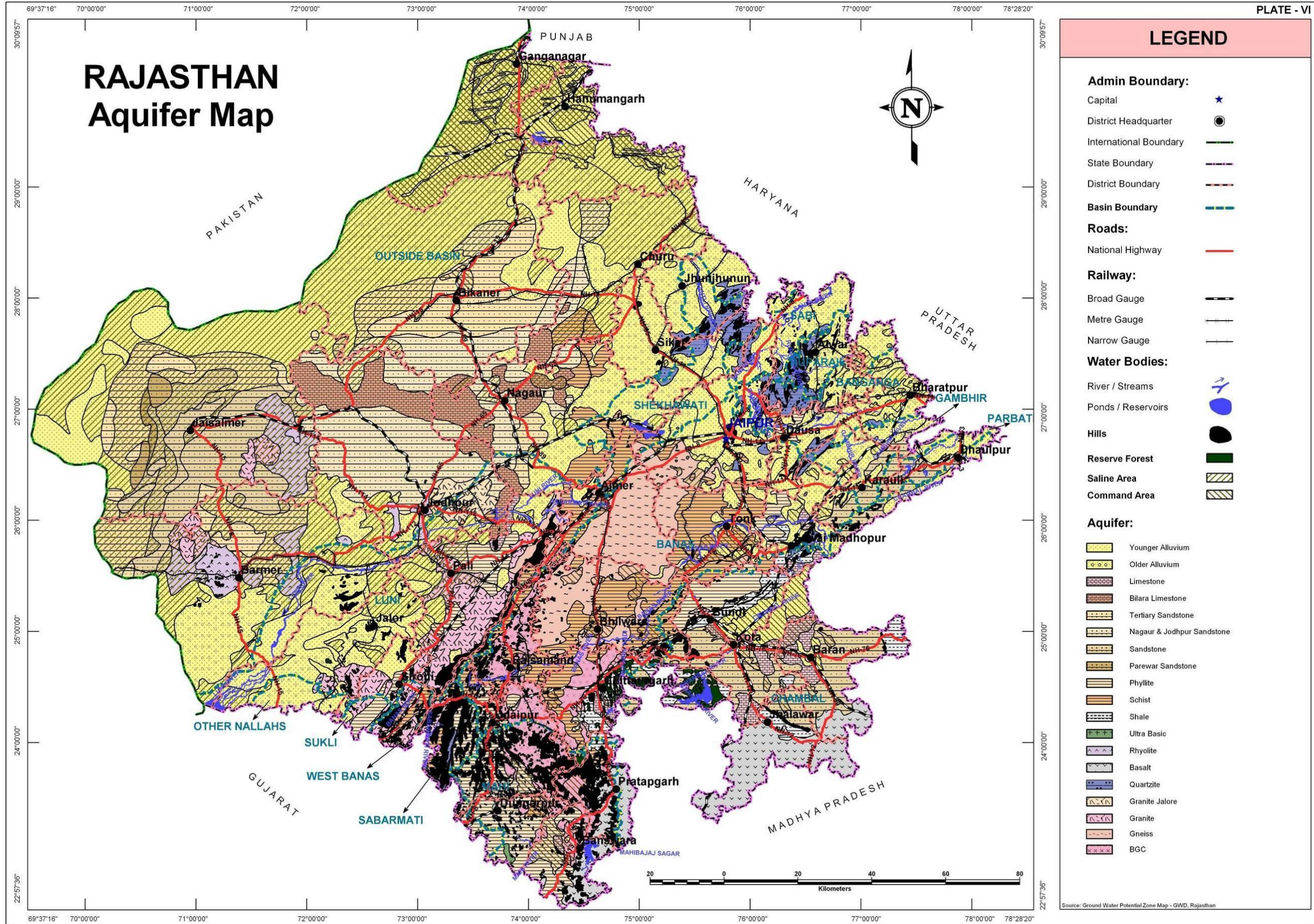


AQUIFERS

Standard Aquifer Type	Area (sq km)	% of total area	Occurrence in District	Description of aquifer
Younger Alluvium	47,662.80	13.92	Ajmer, Baran, Barmer, Bundi, Churu, Dausa, Ganganagar, Hanumangarh, Jaipur, Jaisalmer, Jalor, Jhunjhunun, Jodhpur, Kota, Pali, Sikar, Sirohi.	It is largely constituted of Aeolian and Fluvial sand, silt, clay, gravel and pebbles in varying proportions.
Older Alluvium	95,134.95	27.78	Alwar, Barmer, Bharatpur, Bikaner, Churu, Dausa, Dhaulpur, Ganganagar, Hanumangarh, Jaipur, Jalor, Jhunjhunun, Jodhpur, Karauli, Nagaur, Pali, Sawai Madhopur, Sikar, Tonk.	This litho unit comprises of mixture of heterogeneous fine to medium grained sand, silt and kankar.
Limestone	4,136.09	1.21	Banswara, Baran, Bhilwara, Bundi, Chittorgarh, Kota, Pali, Sawai Madhopur.	In general, it is fine to medium grained, grey, red yellowish, pink or buff in colour.
Bilara Limestone	9,102.33	2.66	Bikaner, Churu, Jodhpur, Nagaur.	It is grey to buff coloured hard and compact.
Tertiary Sandstone	24,272.32	7.09	Barmer, Bikaner, Jaisalmer, Nagaur.	Medium to coarse grained, consolidated to semi consolidated sandstone.
Nagaur & Jodhpur Sandstone	23,920.14	6.99	Bikaner, Churu, Jaisalmer, Jodhpur, Nagaur.	Buff to reddish brown in colour, fine to medium grained hard and compact sandstone.
Vindhyan Sandstone	26,930.64	7.87	Baran, Bhilwara, Bundi, Chittorgarh, Dhaulpur, Jhalawar, Karauli, Kota.	Fine to medium grained, red colour and compact and at places.
Parewar Sandstone	1,285.44	0.38	Jaisalmer.	It is feldspathic ferruginous sandstone.
Gneiss	13,776.11	4.02	Ajmer, Bhilwara, Dausa, Jaipur, Pali, Tonk.	Comprises of porphyritic and non porphyritic gneissic complex.
Phyllite	14,456.52	4.22	Banswara, Bundi, Dausa, Dungarpur, Pali, Rajsamand, Sawai Madhopur, Sirohi, Udaipur.	These include meta sediments and represented by carbonaceous phyllite.
Schist	21,438.73	6.26	Ajmer, Bhilwara, Chittorgarh, Churu, Dungarpur, Jaipur, Jodhpur, Nagaur, Pratapgarh, Rajsamand, Sirohi, Tonk, Udaipur.	Medium to fine grained compact rock. The litho units are soft, friable and have closely spaced cleavage.
Shale	4,574.00	1.34	Baran, Bundi, Chittorgarh, Jhalawar, Kota, Pratapgarh, Sawai Madhopur.	Grey, light green and purple in colour and mostly splintery in nature.
Ultra Basic	103.28	0.03	Dungarpur.	This comprises serpentinite, hyperstinite and amphibolite.
Rhyolite	6,588.26	1.92	Barmer, Jaisalmer, Jalor, Jodhpur.	Rhyolite is porphyritic and has phenocryst of quartz and feldspar.
Basalt	8,969.64	2.62	Banswara, Baran, Jhalawar, Pratapgarh.	Dark grey, olive green and green colour, compact, vesicular, amygdaloidal and weathered.
Quartzite	4,998.73	1.46	Alwar, Bharatpur, Dausa, Jaipur, Jhunjhunun, Karauli, Sawai Madhopur, Sikar, Udaipur.	Medium to coarse grained and varies from feldspathic grit to sericitic quartzite.
Granite Jalore	1,852.64	0.54	Jodhpur.	Grey to pink in colour, medium to coarse grained, and non porphyritic.
Granite	9,431.08	2.75	Barmer, Jaipur, Jaisalmer, Jalor, Pali, Sirohi, Udaipur.	Light grey to pink colour, medium to coarse grained, and characteristically have porphyritic texture.
BGC	8,324.08	2.43	Banswara, Chittorgarh, Dungarpur, Pratapgarh, Rajsamand, Udaipur.	Grey to dark coloured, medium to coarse grained rocks.
Hills	14,058.86	4.11	Ajmer, Alwar, Banswara, Baran, Barmer, Bharatpur, Bhilwara, Bundi, Chittorgarh, Dausa, Dungarpur, Hanumangarh, Jaipur, Jalor, Jhalawar, Jhunjhunun, Jodhpur, Karauli, Nagaur, Pali, Pratapgarh, Rajsamand, Sawai Madhopur, Sikar, Sirohi, Tonk, Udaipur.	-
Reserve Forest	1,387.87	0.41	Chittorgarh, Jalor.	-
TOTAL	342,404.50	100.00		



PLATE - VI



CATEGORIZATION: STAGE OF GROUND WATER DEVELOPMENT

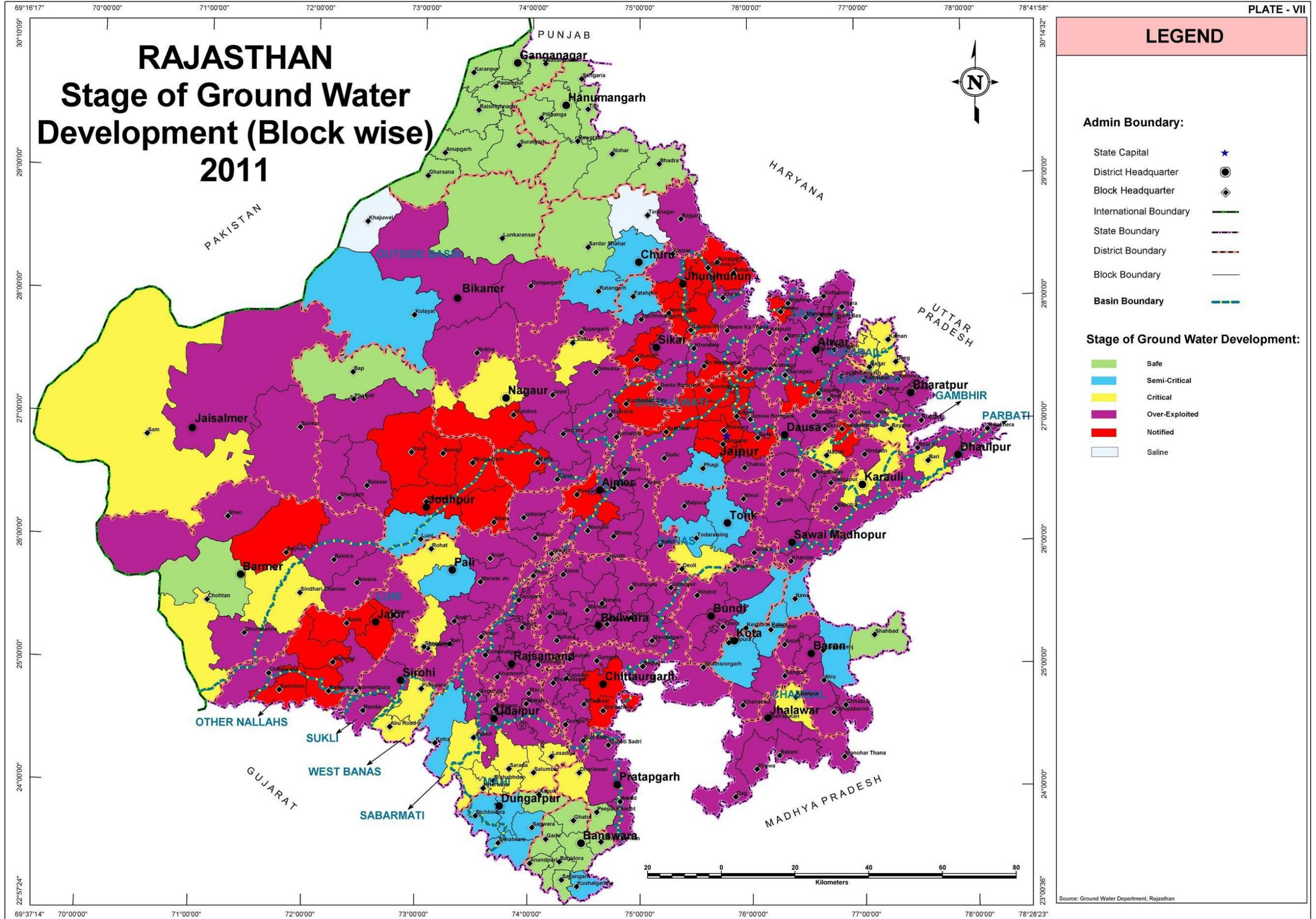
Block wise analysis of development status of ground water when categorized into safe to over exploited stage indicates that 55% of all the blocks are currently being over exploited whereas only 15% of the blocks fall under safe category. About 13% of the blocks have been notified where any further development of ground water is not permitted.

Categorization on the basis of development of GW	%age of State	Block Name
Safe	13.33%	Anandpuri, Anupgarh, Aspur, Bagidora, Banswara, Bap, Barmer, Bhadra, ChhotiSarwan, Ganganagar, Garhi, Gharsana, Ghatol, Hanumangarh, Karanpur, Lunkaransar, Nohar, Padampur, Peepalkhoont, Pilibanga, Raisinghnagar, Rawatsar, Sadulshahar, Sajjangarh, Sangaria, Sardar Shahar, Shahbad, Suratgarh, Tibi
Semi-Critical	9.05%	Bichhiwara, Churu, Dungarpur, Fatehpur, Itawa, Keshorai Patan, Khajuwat, Kishanganj, Kolayat, Kotra, Kushalgarh, Ladpura, Luni, Pali, Phagi, Ratangarh, Sagwara, Simalwara, Sultanpur, Todaraingh, Tonk
Critical	14.46%	Sindhari, Chohtan, Kaman, Deeg, Bayana, Nagar, Bari, Sam, Khanpur, Nadoti, Karauli, Nagaur, Ladnu, Rohat, Sumerpur, Dhariawad, Pindwara, Abu Road, Deoli, Kherwara, Rishabhdeo, Jhadol, Sarada, Lasadiya, Salumbar.
Over Exploited	50.40%	Ahore, Alsisar, Amet, Antah, Arain, Arnod, Asind, Atru, Bakani, Balesar, Bali, Balotra, Bamanwas, Bandikui, Banera, Bansur, Baran, Bargaon, Bari Sadri, Baseri, Begun, Bhadesar, Bhainsrorgarh, Bhim, Bhinay, Bhinder, Bhopalsagar, Bikaner, Bonli, Bundi, Chaksu, Chhabra, Chhipabarod, Chhoti Sadri, Chitalwana, Dag, Danta Ramgarh, Dausa, Degana, Deogarh, Desuri, Dhaulpur, Dhorimanna, Didwana, Dudu, Dungargarh, Dungla, Gangapur, Gangrar, Girwa, Gogunda, Hindaun, Hindoli, Hurda, Jahazpur, Jaisalmer, Jaitaran, Jamwa Ramgarh, Jaswantpura, Jawaja, Jayal, Jhalrapatan, Kapasan, Kathumar, Kekri, Khairabad, Khamnor, Khandar, Khandela, Khetri, Kishanganj, Kishanganj Bas, Kotkasim, Kotputli, Kotri, Kumbhalgarh, Kumher, Lachhmangarh, Lalsot, Mahwa, Makrana, Malpura, Mandal, Mandalgarh, Mandawar, Manohar Thana, Marwar Junction, Masuda, Mavli, Nadbai, Nainwa, Neem Ka Thana, Neemrana, Niwai, Nokha, Parbatsar, Phalodi, Piprali, Pirawa, Pratapgarh, Railmagra, Raipur, Rajakhera, Rajgarh, Rajsamand, Ramgarh, Rani, Rashmi, Reni, Reodar, Riyan, Rupbas, Sahara, Sangod, Sankra, Sapotra, Sawai Madhopur, Sewar, Shahpura, Sheo, Sheoganj, Shergarh, Sikrai, Sirohi, Siwana, Sojat, Srinagar, Sujangarh, Suwana, Talera, Taranagar, Thanagazi, Tijara, Umren, Uniara, Viratnagar, Weir
Over Exploited (Notified)	11.66%	Amber, Bassi, Bawari, Baytoo, Behror, Bhinmal, Bhopalgarh, Bilara, Buhana, Chirawa, Chittaurgarh, Dhond, Govindgarh, Jalor, Jhotwara, Jhunjhun, KuchamanCity, Mandor, Merta, Mundwa, Nawalgarh, Nimbahera, Osian, Peesangan, Rajgarh, Raniwara, Sambhar, Sanchore, Sanganer, Sayla, Shahpura, Srimadhapur, Surajgarh, Todabhim, Udaipurwati
Saline	1.10%	Khajuwat, Taranagar

Basis for categorization:

Ground water development	Categorization
<= 70%	Safe
70% - 90%	Semi critical
90% - 100%	Critical
> 100%	Over exploited

In **Notified Blocks**, development of ground water is not permitted any more.

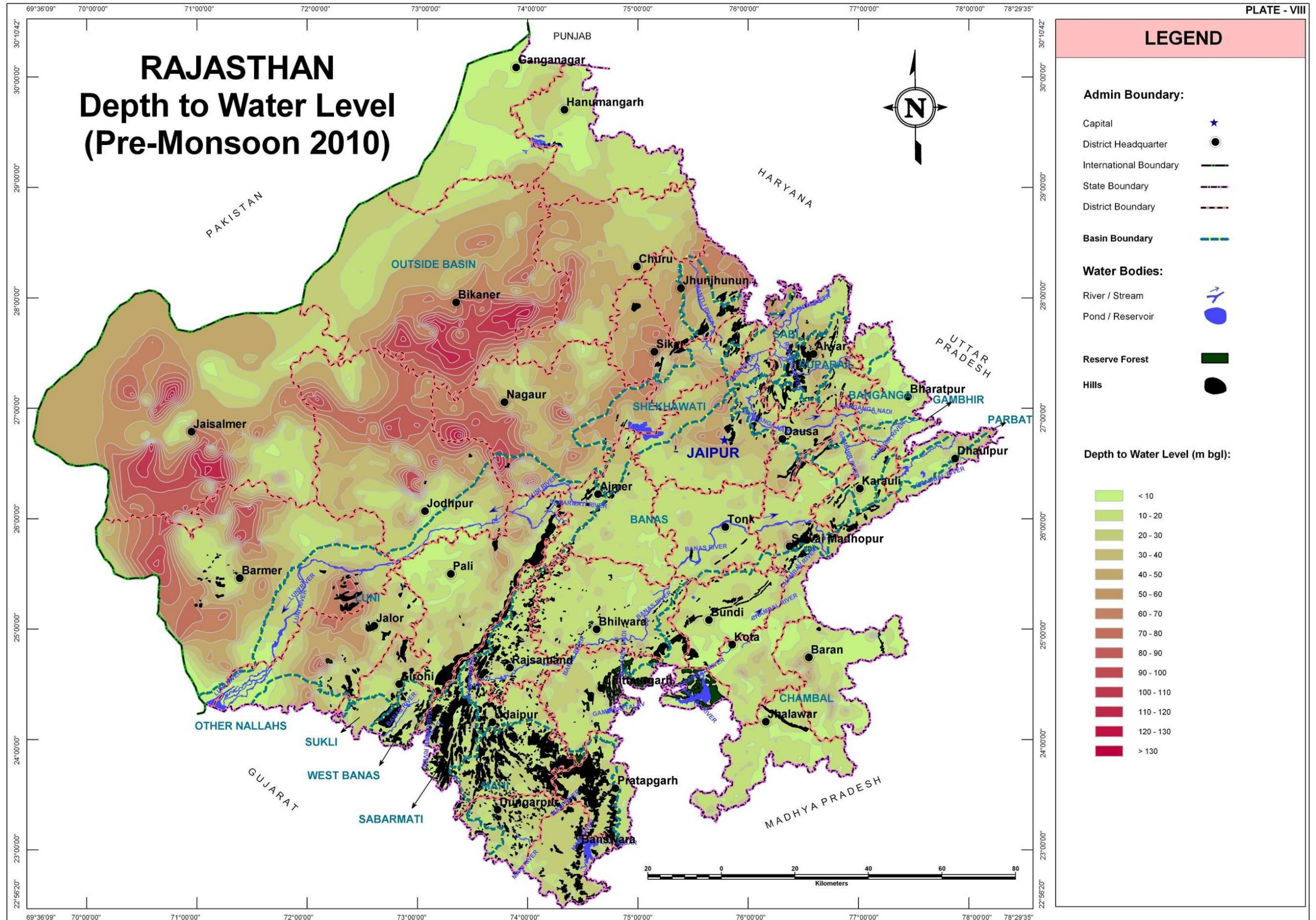


DEPTH TO WATER LEVEL (PRE MONSOON – 2010)

Depth to water level in the state varies from less than 10m below ground level to more than 130m. A quick comparison of this map with rainfall distribution reveals a somewhat direct relationship, i.e., the areas that receive better rainfall have groundwater occurring at shallower depths whereas in rain deficient areas water occurs at far greater depths. In addition, around major cities the water has generally gone down as seen around most of the district headquarters. East of Aravalli, the general depth to water level is less than 30m, except around major cities; and in the western part most of the area shows deeper water table i.e., more than 50m and reaching upto 130m of depth. The shallower water levels in the west is largely restricted to the vicinity of Indira Gandhi Canal and some isolated pockets around Barmer, east of Jaisalmer and around Pali.

District Name	Area (in sq km) in each district within depth to water level range (Pre Monsoon – 2010, m bgl) *														Total
	< 10m	10-20m	20-30m	30-40m	40-50m	50-60m	60-70m	70-80m	80-90m	90-100m	100-110m	110-120m	120-130m	> 130m	
Ajmer	398.3	5,218.5	2,166.8	418.3	11.8	-	-	-	-	-	-	-	-	-	8,213.7
Alwar	24.8	2,532.3	2,588.8	1,368.9	626.0	281.9	37.9	-	-	-	-	-	-	-	7,460.5
Banswara	60.9	668.1	3,338.0	6.5	-	-	-	-	-	-	-	-	-	-	4,073.5
Baran	3,110.9	2,672.2	707.3	334.0	116.0	25.5	-	-	-	-	-	-	-	-	6,965.9
Barmer	1,039.4	4,046.5	6,074.7	3,838.7	3,378.7	2,564.3	3,124.4	2,149.2	1,354.2	668.2	41.0	-	-	-	28,279.3
Bharatpur	1,901.2	2,624.6	533.8	19.2	-	-	-	-	-	-	-	-	-	-	5,078.9
Bhilwara	371.4	7,597.7	1,776.5	73.5	7.9	-	-	-	-	-	-	-	-	-	9,827.0
Bikaner	216.5	3,200.1	6,295.3	647.6	5,982.1	3,384.0	2,648.6	1,971.6	1,962.2	1,944.8	1,422.5	557.1	46.8	-	30,279.2
Bundi	1,458.0	3,366.7	764.2	2.0	-	-	-	-	-	-	-	-	-	-	5,590.9
Chittorgarh	256.0	4,133.3	1,883.6	110.4	3.9	-	-	-	-	-	-	-	-	-	6,387.2
Churu	4.7	1,308.8	3,361.4	2,580.4	3,882.4	1,573.9	423.1	312.1	366.5	29.0	1.8	-	-	-	13,844.2
Dausa	-	448.4	1,680.5	992.9	104.7	0.5	-	-	-	-	-	-	-	-	3,227.0
Dhaulpur	607.0	1,653.2	559.8	220.3	0.0	-	-	-	-	-	-	-	-	-	3,040.3
Dungarpur	69.9	1,529.9	1,885.5	5.3	-	-	-	-	-	-	-	-	-	-	3,490.6
Ganganagar	4,086.8	4,900.3	1,182.9	-	514.1	-	-	-	-	-	-	-	-	-	10,684.0
Hanumangarh	1,508.7	5,412.4	2,079.2	0.0	869.5	39.9	-	-	-	-	-	-	-	-	9,909.7
Jaipur	15.2	2,566.5	3,506.4	2,775.0	1,330.2	477.4	96.2	18.6	-	-	-	-	-	-	10,785.5
Jaisalmer	476.7	4,088.9	4,628.6	821.5	13,826.5	4,386.7	3,335.1	2,128.9	2,232.4	1,454.9	687.8	352.0	61.4	5.8	38,487.2
Jalor	1,409.8	1,489.2	2,919.0	1,504.8	1,464.6	996.9	445.0	165.4	28.3	4.4	-	-	-	-	10,427.4
Jhalawar	1,058.3	4,860.7	261.7	24.5	0.2	-	-	-	-	-	-	-	-	-	6,205.4
Jhunjhun	-	0.7	333.0	1,468.5	1,260.9	1,541.2	999.4	29.6	-	-	-	-	-	-	5,633.3
Jodhpur	1,030.8	2,787.9	3,597.1	3,403.1	2,853.3	2,115.7	2,357.5	2,545.1	1,455.8	489.0	56.6	5.1	-	-	22,697.0
Karauli	78.0	2,431.7	1,787.5	237.9	90.3	31.2	14.2	1.0	-	-	-	-	-	-	4,671.8
Kota	1,203.1	2,492.3	1,258.1	146.7	22.1	-	-	-	-	-	-	-	-	-	5,122.3
Nagaur	35.9	495.9	2,045.4	3,634.4	3,999.1	2,714.7	1,559.9	1,207.0	1,279.0	424.0	191.1	124.6	53.5	6.1	17,770.6
Pali	1,440.4	3,891.4	3,572.7	1,564.0	562.3	233.4	0.6	-	-	-	-	-	-	-	11,264.8
Pratapgarh	29.1	1,449.5	1,445.5	-	-	-	-	-	-	-	-	-	-	-	2,924.1
Rajsamand	12.1	3,148.8	618.2	10.8	-	-	-	-	-	-	-	-	-	-	3,789.9
Sawai Madhopur	208.6	2,409.3	1,707.9	255.2	75.3	15.9	-	-	-	-	-	-	-	-	4,672.2
Sikar	0.1	355.0	1,209.0	1,799.4	1,780.2	1,309.7	875.6	44.4	-	-	-	-	-	-	7,373.4
Sirohi	37.8	1,996.4	1,851.6	379.7	70.4	-	-	-	-	-	-	-	-	-	4,335.9
Tonk	168.4	6,056.0	916.1	10.6	-	-	-	-	-	-	-	-	-	-	7,151.1
Udaipur	339.4	3,978.8	2,901.6	76.7	7.0	-	-	-	-	-	-	-	-	-	7,303.5
Total	22,658.2	95,812.0	71,437.7	28,730.8	42,839.50	21,692.8	15,917.5	10,572.9	8,678.4	5,014.3	2,400.8	1,038.8	161.7	11.9	326,967.3

* Area covered in the derived map is less than the total state area since the hills and reserve forest have been excluded from interpolation/contouring



WATER TABLE ELEVATION (PRE MONSOON – 2010)

Water table elevation map provide a different perspective and groundwater flow directions can be visualized from this map. Aravallis act as regional flow barriers and ground water flows towards east to the east of this range and to the west in the western part. In Mt. Abu area, maximum water table elevation of about 920m amsl was observed and minimum elevation of less than 40 m amsl is found in the western and southwestern part of the state bordering with Pakistan and Gujarat.

District	Area (in sq km) in each district within water table elevation range (Pre Monsoon – 2010, m amsl)															Total
	< 40	40 - 80	80 - 120	120 - 160	160 - 200	200 - 240	240 - 280	280 - 320	320 - 360	360 - 400	400 - 440	440 - 480	480 - 520	520 - 560	560 - 920	
Ajmer	-	-	-	-	-	-	-	336.9	1,488.8	2,742.3	2,114.8	1,115.2	274.4	83.1	58.2	8,213.7
Alwar	-	-	-	-	329.7	2,171.3	2,633.2	1,017.6	649.9	445.1	213.3	0.4	-	-	-	7,460.5
Banswara	-	-	1.8	546.2	1,025.4	1,101.5	482.1	361.0	246.4	273.0	36.0	-	-	-	-	4,073.5
Baran	-	-	-	-	15.2	717.4	1,913.1	1,190.9	1,705.7	912.9	479.0	31.8	-	-	-	6,965.9
Barmer	2,809.0	6,421.7	7,999.7	5,501.3	3,941.9	1,502.7	92.1	10.9	-	-	-	-	-	-	-	28,279.3
Bharatpur	-	-	-	-	4,217.9	861.0	-	-	-	-	-	-	-	-	-	5,078.9
Bhilwara	-	-	-	-	-	-	1.1	296.5	2,401.3	2,430.0	1,795.7	1,564.9	952.4	366.6	18.5	9,827.0
Bikaner	-	740.8	3,453.5	16,457.8	5,411.6	3,600.3	615.2	-	-	-	-	-	-	-	-	30,279.2
Bundi	-	-	-	-	11.3	1,570.8	2,514.6	861.6	180.4	148.9	168.5	122.7	12.1	-	-	5,590.9
Chittorgarh	-	-	-	-	-	-	10.0	424.6	377.5	1,228.4	2,593.6	1,471.5	217.7	63.9	-	6,387.2
Churu	-	-	-	645.3	3,324.0	4,479.6	3,638.8	1,756.4	-	-	-	-	-	-	-	13,844.2
Dausa	-	-	-	-	20.7	637.8	1,275.7	1,292.8	0.0	-	-	-	-	-	-	3,227.0
Dhaulpur	-	-	-	1,001.7	752.1	796.1	468.4	22.0	-	-	-	-	-	-	-	3,040.3
Dungarpur	-	-	20.8	575.0	1,006.6	719.0	566.3	453.5	149.6	-	-	-	-	-	-	3,490.6
Ganganagar	-	-	-	7,581.0	3,102.9	-	-	-	-	-	-	-	-	-	-	10,684.0
Hanumangarh	-	-	-	1,726.8	8,087.2	95.7	-	-	-	-	-	-	-	-	-	9,909.7
Jaipur	-	-	-	-	-	-	154.0	2,586.4	2,852.9	3,034.5	1,861.5	296.2	-	-	-	10,785.5
Jaisalmer	4,285.1	9,111.0	7,459.2	8,540.8	4,394.1	3,049.5	1,333.9	313.4	-	-	-	-	-	-	-	38,487.2
Jalor	2,884.4	2,326.0	989.1	1,786.1	1,902.1	410.5	129.0	-	-	-	-	-	-	-	-	10,427.4
Jhalawar	-	-	-	-	-	-	264.7	1,274.7	2,145.8	1,687.3	832.9	-	-	-	-	6,205.4
Jhunjhunun	-	-	-	-	33.9	1,163.4	1,603.7	1,097.3	808.9	668.4	243.3	14.5	-	-	-	5,633.3
Jodhpur	-	-	-	1106.9	9665.9	9,738.2	2186.0	-	-	-	-	-	-	-	-	22,697.0
Karauli	-	-	-	-	173.9	2,387.9	1,590.5	496.6	22.9	-	-	-	-	-	-	4,671.8
Kota	-	-	-	-	99.2	1,734.3	1,453.5	1,174.0	340.1	312.7	8.4	-	-	-	-	5,122.3
Nagaur	-	-	-	-	207.7	3,500.3	5,486.4	3,750.2	2,269.2	2,059.6	476.3	20.7	-	-	-	17,770.6
Pali	-	-	-	113.2	1,678.8	3,674.2	2,657.0	1,484.5	928.0	457.9	190.5	61.6	14.9	4.1	-	11,264.8
Pratapgarh	-	-	-	-	47.3	340.2	205.1	115.2	120.6	160.1	234.6	918.4	724.0	58.7	-	2,924.1
Rajsamand	-	-	-	-	-	-	-	-	-	-	0.3	397.0	548.5	751.6	2,092.6	3,789.9
Sawai Madhopur	-	-	-	-	183.1	2,207.7	2,208.4	72.4	0.6	-	-	-	-	-	-	4,672.2
Sikar	-	-	-	-	-	-	539.0	1,316.6	964.7	1,335.1	2,094.0	1,122.4	1.5	-	-	7,373.4
Sirohi	-	-	-	-	344.6	1,175.6	1,208.5	754.5	430.6	321.0	101.2	-	-	-	-	4,335.9
Tonk	-	-	-	-	-	34.5	2,589.8	3,561.6	935.7	29.5	-	-	-	-	-	7,151.1
Udaipur	-	-	-	-	135.0	388.1	496.7	820.2	542.1	468.5	474.6	1,125.5	854.9	576.0	1,422.0	7,303.5
Total	9,978.5	18,599.5	19,924.1	4,5582.2	5,0112.1	48,057.6	38,317.0	26,842.5	19,561.5	18,715.3	13,918.6	8,262.7	3,600.4	1,904.0	3,591.3	326,967.3



PLATE - IX

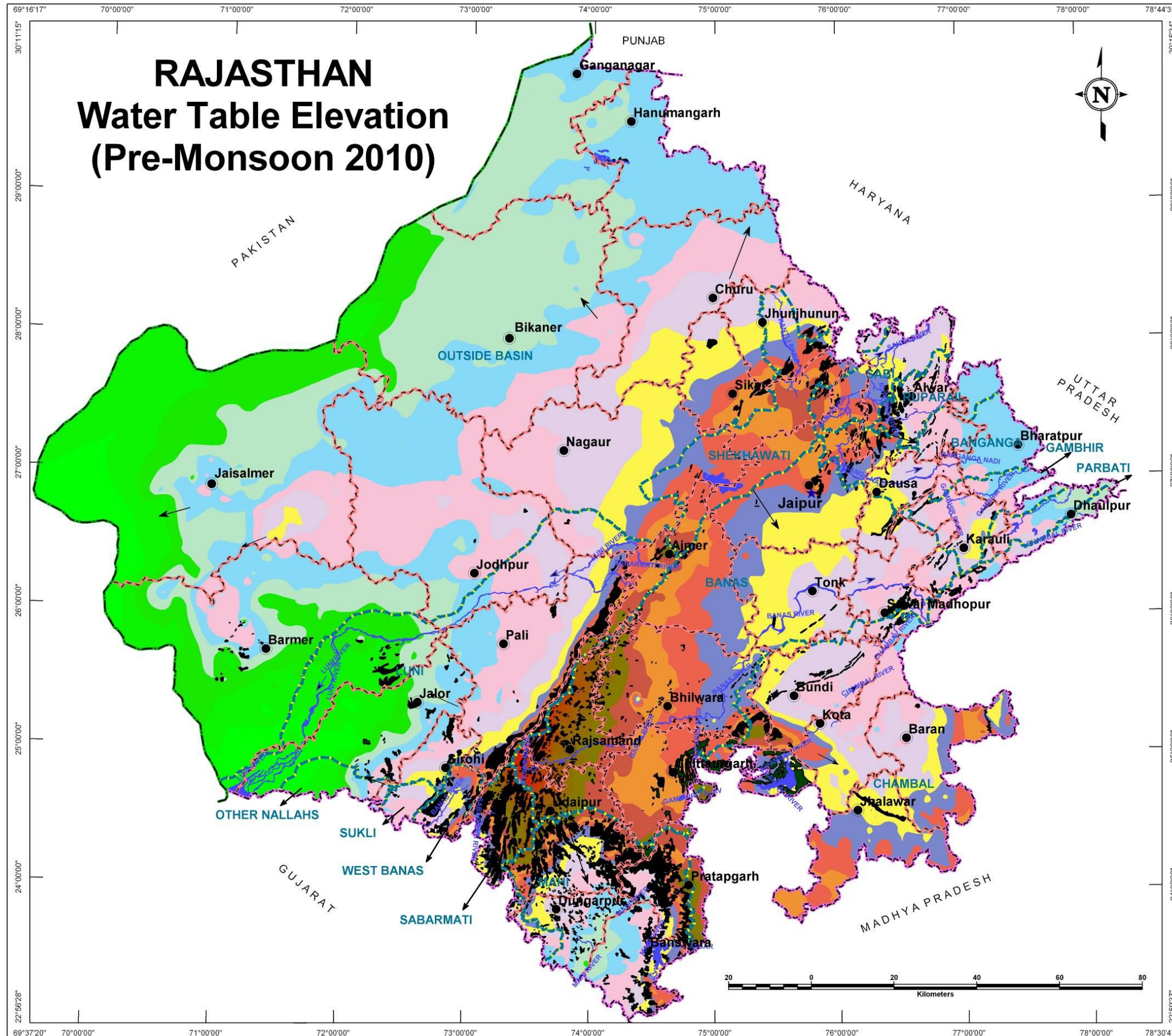
LEGEND

Admin Boundary:

- State Capital
- District Headquarter
- International Boundary
- State Boundary
- District Boundary
- Basin Boundary
- Water Bodies:
 - River / Streams
 - Ponds / Reservoirs
- Hills
- Reserve Forest
- Water Table Elevation with Ground Water Flow Direction

Water Table Elevation (m amsl):

- < 40
- 40 - 80
- 80 - 120
- 120 - 160
- 160 - 200
- 200 - 240
- 240 - 280
- 280 - 320
- 320 - 360
- 360 - 400
- 400 - 440
- 440 - 480
- 480 - 520
- 520 - 560
- 560 - 600
- 600 - 640
- 640 - 680
- 680 - 720
- 720 - 760
- 760 - 800
- 800 - 840
- 840 - 880
- 880 - 920
- > 920



WATER LEVEL FLUCTUATION (PRE TO POST MONSOON 2010)

Difference in water level between pre-monsoon and post monsoon measurements bring forth areas that had shown increase or decrease in ground water storage, by the end of monsoon period. The –ve fluctuation areas indicate that water level has gone down in post monsoon season i.e., ground water in those areas are being withdrawn from static ground water resources and the situation is alarming. Such situation persists in major part of western Rajasthan and in pockets north of Jaipur and around Alwar, Kota, Baran etc. Significant decline of water table by 6-8m and more is indicated near Bikaner, Barmer, Hanumangarh, Karauli etc. However, a general rise of water level by 2m and upto 6m is observed over the larger part of the areas in the state, and some localized pockets have shown significant rise of more than 10 m and upto 18m west of Bhilwara, Bundi, Tonk and adjacent to the Aravalli range, largely in hardrock aquifers.

District	Area (sq km) within WL Fluctuation range (m)															Total
	<-8 (m)	(-8) to (-6) (m)	(-6) to (-4) (m)	(-4) to (-2) (m)	(-2) to 0 (m)	0 to 2 (m)	2 to 4 (m)	4 to 6 (m)	6 to 8 (m)	8 to 10 (m)	10 to 12 (m)	12 to 14 (m)	14 to 16 (m)	16 to 18 (m)	> 18 (m)	
Ajmer	-	-	3.5	16.7	49.7	471.2	1,308.9	1,915.1	2,116.7	1,360.1	596.1	286.3	60.2	22.7	6.3	8,213.7
Alwar	-	0.8	2.8	25.6	747	3,787.60	1,940.3	729.6	189.4	23.2	10.6	3.6	-	-	-	7,460.5
Banswara	-	-	-	-	6.4	1,724.50	1,189.4	721.4	275.0	85.1	38.0	17.8	10.9	4.8	-	4,073.5
Baran	-	3.5	9.4	114.1	1,043.60	2,631.70	2,122.3	727.2	214.0	63.2	30.5	5.9	0.4	-	-	6,965.9
Barmer	-	-	31.5	238.7	5,305.30	15,058.80	5,154.7	1,758.4	460.7	153.4	69.7	35.9	12.1	-	-	28,279.3
Bharatpur	-	-	-	-	34.5	2,129.40	2,257.7	494.8	117.3	31.2	13.3	0.6	-	-	-	5,078.9
Bhilwara	-	-	3.7	18.1	64.7	348.9	1,301.8	1,841.4	2,108.5	1,881.5	1,184.0	668.9	234.5	106.0	65.1	9,827.0
Bikaner	-	78.4	168.4	617.7	15,585.20	13,188.50	482.8	152.2	6.0	-	-	-	-	-	-	30,279.2
Bundi	-	-	-	57.9	448.1	564.9	803.5	1,214.8	1,027.0	761.4	257.0	181.4	144.0	116.5	14.3	5,590.9
Chittorgarh	-	-	-	-	18.0	386.9	1,136.0	1,545.6	1,332.2	1,028.3	661.9	235.6	40.8	1.9	-	6,387.2
Churu	-	-	-	36.6	5,750.40	7,815.30	125.0	49.0	29.8	18.4	10.9	6.0	2.4	0.5	-	13,844.2
Dausa	-	-	-	34.7	560.8	1,348.70	817.7	286.2	68.2	41.7	44.6	18.4	6.1	-	-	3,227.0
Dhaulpur	0.2	2.8	30.1	41.5	246.4	883.4	721.4	618.6	411.2	79.8	4.9	-	-	-	-	3,040.3
Dungarpur	-	-	-	-	3.9	535.8	1,275.5	1,079.3	444.8	114.6	25.9	7.5	3.3	-	-	3,490.6
Ganganagar	-	-	-	67.7	4,872.90	5,144.80	483.2	96.2	15.3	3.8	-	-	-	-	-	10,684.0
Hanumangarh	-	4.4	91.0	597.9	5,224.00	3,480.00	404.7	91.7	16.0	-	-	-	-	-	-	9,909.7
Jaipur	-	-	6.1	58.0	1,431.20	3,682.50	2,161.5	1,465.8	779.9	658.8	361.6	102.7	57.0	16.9	3.4	10,785.5
Jaisalmer	-	-	64.3	424.8	9,290.40	24,333.60	3,138.8	867.7	243.4	66.5	39.5	18.2	-	-	-	38,487.2
Jalor	-	1.2	16.3	80.5	979.3	5,365.30	2,142.1	898.7	453.7	246.1	102.2	68.7	73.3	-	-	10,427.4
Jhalawar	-	4.1	14.0	45.2	164.8	1,534.70	2,746.4	1,257.2	307.7	100.9	24.6	4.1	1.3	0.2	-	6,205.4
Jhunjhun	-	-	-	-	1,429.00	2,735.00	638.5	370.7	189.1	101.6	63.8	34.8	28.5	36.8	5.5	5,633.3
Jodhpur	-	-	-	24.4	13,282.80	1,781.20	4,925.9	1,844.8	475.6	218.9	109.8	28.1	3.5	1.9	0.1	22,697.0
Karauli	-	2.8	28.0	71.5	238	1,702.60	1,566.1	561.9	231.9	126.0	74.7	41.5	19.2	6.8	0.7	4,671.8
Kota	-	2.3	6.9	88.1	782.1	1,261.10	900.6	607.8	470.7	343.8	356.0	185.0	103.3	14.6	-	5,122.3
Nagaur	-	-	-	1.0	5,120.40	11,720.50	631.3	164.1	79.4	28.0	13.3	7.8	4.3	0.4	-	17,770.6
Pali	0.1	2.9	8.6	23.7	92	1,722.00	3,360.2	2,366.3	1,519.2	1,057.8	599.5	410.9	87.9	13.6	0.1	11,264.8
Pratapgarh	-	-	-	3.4	67.4	936.8	1,194.7	560.0	159.5	0.4	-	-	-	-	-	2,924.1
Rajsamand	-	-	-	-	-	20.7	104.6	384.5	710.2	1,220.7	875.4	376.5	92.6	4.8	-	3,789.9
Sawai Madhopur	-	2.0	24.3	40.7	43.5	817.8	1,963.9	648.1	413.4	380.9	196.5	82.7	43.4	13.5	1.7	4,672.2
Sikar	-	3.8	17.2	86.5	3,614.40	2,464.10	311.6	196.1	205.8	167.4	150.8	118.2	25.5	10.3	1.8	7,373.6
Sirohi	-	-	-	-	8	255.5	554.0	854.7	1,070.8	1,067.1	258.4	148.1	77.7	36.5	5.2	4,335.9
Tonk	-	-	-	-	62	273.1	1,446.0	1,935.4	1,667.1	985.3	582.0	156.2	38.0	6.2	-	7,151.1
Udaipur	-	0.4	2.9	7.6	80.9	191	1,730.4	1,671.8	1,478.2	1,066.4	685.9	310.9	75.0	7.2	0.7	7,309.4
Total	0.2	109.3	529.1	2,823.0	65,145.40	131,800.40	51,041.2	29,975.0	19,283.2	13,475.0	7,441.7	3,561.9	1,245.1	422.2	105.0	326,967.3



PLATE - X

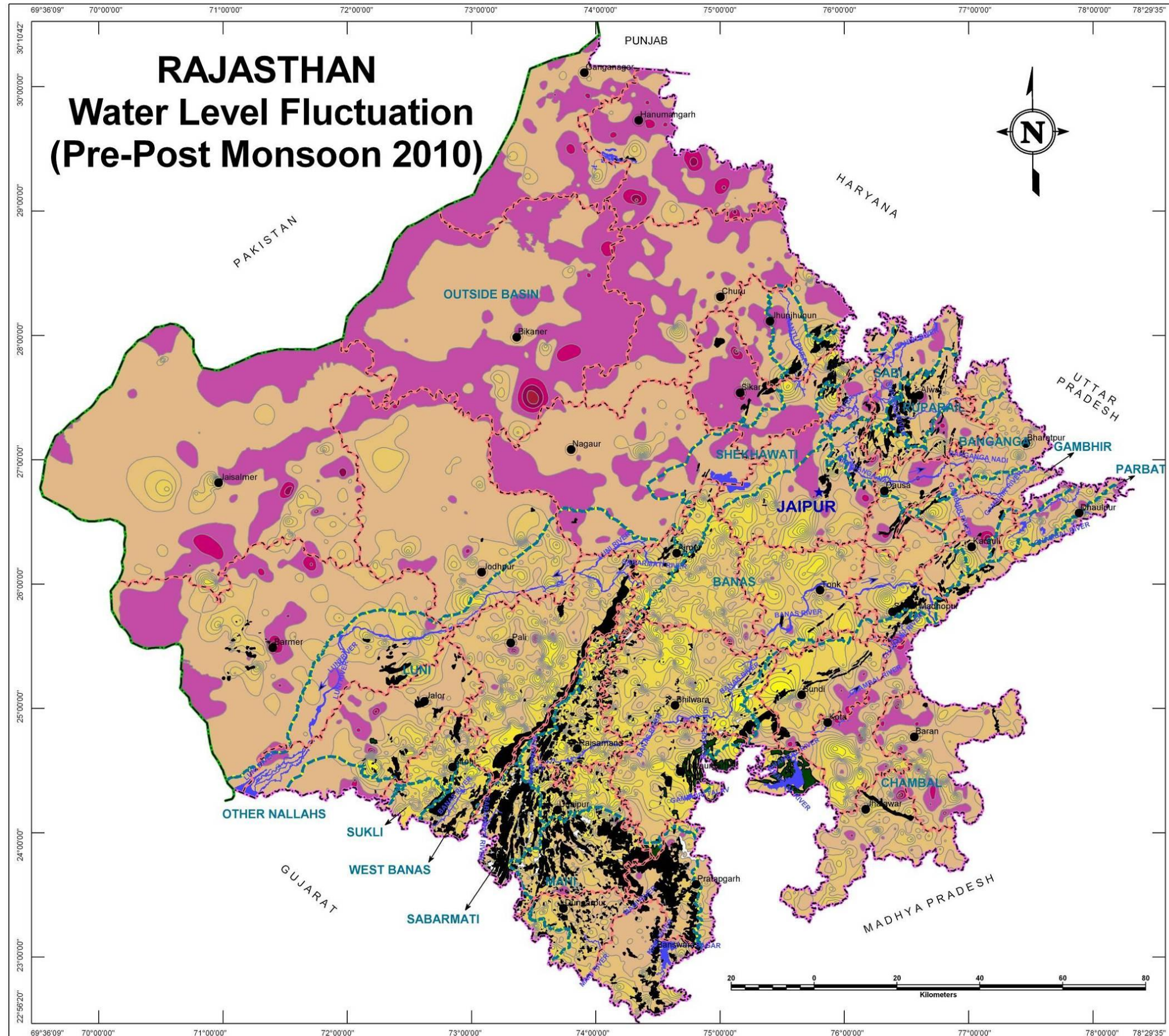
LEGEND

Admin Boundary:

- Capital: ★
- District Headquarter: ●
- International Boundary: —
- State Boundary: - - -
- District Boundary: - . - . -
- Basin Boundary: - - -
- Water Bodies:
- River / Stream: ~
- Pond / Reservoir: ○
- Reserve Forest: ■
- Hills: ●

Water Level Fluctuation (m):

- < -8
- 8 to -6
- 6 to -4
- 4 to -2
- 2 to 0
- 0 to 2
- 2 to 4
- 4 to 6
- 6 to 8
- 8 to 10
- 10 to 12
- 12 to 14
- 14 to 16
- 16 to 18
- > 18

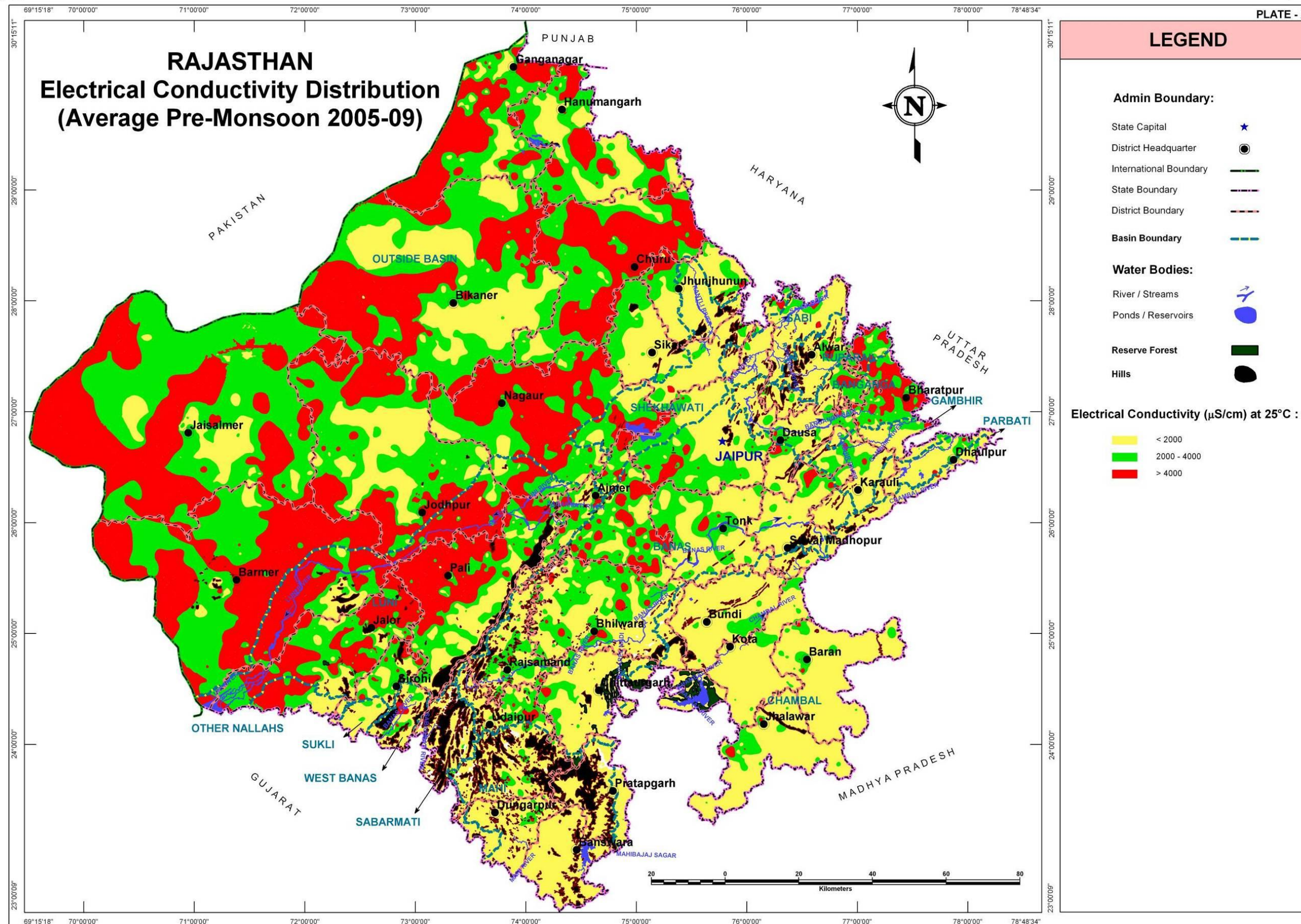


GROUND WATER ELECTRICAL CONDUCTIVITY DISTRIBUTION

High Electrical Conductivity values are seen largely in western Rajasthan especially around Pali, Barmer, Jodhpur, west of Bikaner, Jaisalmer and Churu with some moderately high salinity areas in between. Except for some very high salinity areas around Bharatpur and in the region between Jaipur, Ajmer, Tonk and Bhilwara; areas in the eastern Rajasthan (around Bundi, Kota, Baran and Jhalawar) and southeastern parts around Banswara and Dungarpur indicate the presence of fresh potable groundwater.

Average values of EC ($\mu\text{S}/\text{cm}$ at 25°C) from measurements between 2005 and 2009 have been plotted and EC ranges selected for depiction are in line with suitability for human drinking purpose.

District Name	<2000 mg/l of EC		2000-4000 mg/l of EC		>4000 mg/l of EC		Total Area
	Area (sq km)	%age of total	Area (sq km)	%age of total	Area (sq km)	%age of total	
Ajmer	2,388	29.07	3,741	45.54	2,085	25.39	8,213.7
Alwar	5,523	74.03	1,508	20.21	430	5.76	7,460.5
Banswara	4,074	100.00	-	0.00	-	0.00	4,073.5
Baran	6,458	92.71	479	6.87	29	0.42	6,965.9
Barmer	1,222	4.32	9,779	34.58	17,278	61.10	28,279.3
Bharatpur	1,176	23.16	2,068	40.71	1,835	36.12	5,078.9
Bhilwara	5,919	60.24	3,429	34.89	479	4.87	9,827.0
Bikaner	8,914	29.44	11,010	36.36	10,355	34.20	30,279.2
Bundi	5,080	90.87	511	9.13	-	0.00	5,590.9
Chittorgarh	5,545	86.87	771	12.08	67	1.05	6,387.2
Churu	1,116	8.06	4,953	35.78	7,774	56.16	13,844.2
Dausa	2,228	69.05	874	27.09	125	3.86	3,227.0
Dhaulpur	2,601	85.56	388	12.78	50	1.66	3,040.3
Dungarpur	3,311	94.85	172	4.94	8	0.22	3,490.6
Ganganagar	1,446	13.53	3,552	33.25	5,686	53.22	10,684.0
Hanumangarh	3,324	33.54	4,002	40.38	2,584	26.07	9,909.7
Jaipur	6,606	61.25	3,618	33.55	561	5.20	10,785.5
Jaisalmer	2,992	7.77	18,205	47.30	17,290	44.92	38,487.2
Jalor	1,569	15.04	3,715	35.63	5,144	49.33	10,427.4
Jhalawar	5,411	87.20	712	11.48	82	1.32	6,205.4
Jhunjhunun	4,119	73.12	1,137	20.18	377	6.70	5,633.3
Jodhpur	5,140	22.66	8,618	37.99	8,929	39.36	22,697.0
Karauli	3,837	82.13	783	16.76	52	1.12	4,671.8
Kota	4,622	90.24	500	9.76	0	0.00	5,122.3
Nagaur	1,991	11.21	6,810	38.32	8,969	50.47	17,770.6
Pali	2,654	23.56	3,435	30.49	5,177	45.96	11,264.8
Pratapgarh	2,900	99.25	22	0.75	-	0.00	2,924.1
Rajsamand	2,423	63.93	1,270	33.50	97	2.57	3,789.9
SawaiMadhopur	3,864	82.71	641	13.72	167	3.57	4,672.2
Sikar	5,230	70.93	1,921	26.05	223	3.02	7,373.4
Sirohi	2,688	61.98	1,330	30.67	319	7.35	4,335.9
Tonk	4,540	63.49	2,388	33.39	223	3.12	7,151.1
Udaipur	6,030	82.49	1,107	15.14	173	2.37	7,303.5
Total	1,26,941	100.0	1,03,449	100.0	96,568	100.0	3,26,967.3



GROUND WATER CHLORIDE DISTRIBUTION

The chloride distribution also follows the similar distribution pattern as that of Electrical conductivity in the State. However, ranges of values selected here for presentation are in line with suitability for drinking purpose by humans based on chloride concentration. Banswara, Bundi, Karauli, Kota, Pratapgarh and to a good extent Sikar, Udaipur, Jhalawar and Jhunjhunun have ground water chloride concentration largely within potability limits. Barmer, Bharatpur, Jaisalmer, Churu etc. have very high chloride concentration in ground water.

Here also, an average chloride concentration of 2005 – 2009 is used for study.

District Name	Chloride <250 mg/l		Chloride 250-1000 mg/l		Chloride >1000 mg/l		Total Area
	Area (sq km)	%age of total	Area (sq km)	%age of total	Area (sq km)	%age of total	
Ajmer	1,301.6	15.8	5,099.6	62.1	1,812.4	22.1	8,213.7
Alwar	4,828.3	64.7	2,297.7	30.8	334.6	4.5	7,460.5
Banswara	4,073.6	100.0	-	0.0	-	0.0	4,073.5
Baran	6,347.4	91.1	614.6	8.8	4.1	0.1	6,965.9
Barmer	328.8	1.2	11,968.7	42.3	15,981.8	56.5	28,279.3
Bharatpur	906.9	17.9	2,461.0	48.5	1,711.0	33.7	5,078.9
Bhilwara	4,033.2	41.0	5,449.9	55.5	343.8	3.5	9,827.0
Bikaner	3,962.5	13.1	17,488.0	57.8	8,828.8	29.2	30,279.2
Bundi	5,116.4	91.5	474.5	8.5	-	0.0	5,590.9
Chittorgarh	4,939.3	77.4	1,426.5	22.3	17.4	0.3	6,387.2
Churu	692.5	5.0	6,732.7	48.6	6,418.8	46.4	13,844.2
Dausa	1,822.2	56.5	1,302.0	40.3	102.7	3.2	3,227.0
Dhaulpur	2,573.0	84.6	462.2	15.2	5.1	0.2	3,040.3
Dungarpur	3,072.2	88.0	416.8	11.9	1.6	0.0	3,490.6
Ganganagar	484.8	4.5	5,139.5	48.1	5,059.7	47.4	10,684.0
Hanumangarh	1,128.7	11.4	6,790.4	68.5	1,990.5	20.1	9,909.7
Jaipur	5,039.9	46.7	5,339.0	49.5	406.4	3.8	10,785.5
Jaisalmer	940.4	2.4	20,517.9	53.3	17,028.8	44.2	38,487.2
Jalor	999.0	9.6	4,383.6	42.0	5,044.9	48.4	10,427.4
Jhalawar	5,294.7	85.3	856.1	13.8	54.6	0.9	6,205.4
Jhunjhunun	3,190.4	56.6	2,321.7	41.2	121.3	2.2	5,633.3
Jodhpur	1,953.5	8.6	13,208.6	58.2	7,524.8	33.2	22,697.0
Karauli	3,429.1	73.4	1,242.8	26.6	-	0.0	4,671.8
Kota	4,761.1	92.9	361.1	7.1	-	0.0	5,122.3
Nagaur	1,357.0	7.6	9,868.8	55.5	6,545.1	36.8	17,770.6
Pali	1,589.2	14.1	4,923.2	43.7	4,752.8	42.2	11,264.8
Pratapgarh	2,434.8	83.3	489.3	16.7	-	0.0	2,924.1
Rajsamand	1,488.3	39.3	2,261.7	59.7	40.0	1.1	3,789.9
Sawai Madhopur	3,483.9	74.6	1,122.6	24.0	65.7	1.4	4,672.2
Sikar	4,577.3	62.1	2,733.8	37.1	62.5	0.8	7,373.4
Sirohi	1,720.6	39.7	2,441.0	56.3	174.3	4.0	4,335.9
Tonk	3,315.0	46.4	3,711.7	51.9	124.3	1.7	7,151.1
Udaipur	5,047.3	69.1	2,187.1	29.9	72.9	1.0	7,303.5
Total	96,233.1	100.0	1,46,094.1	100.0	84,630.6	100.0	3,26,967.3



PLATE - XII

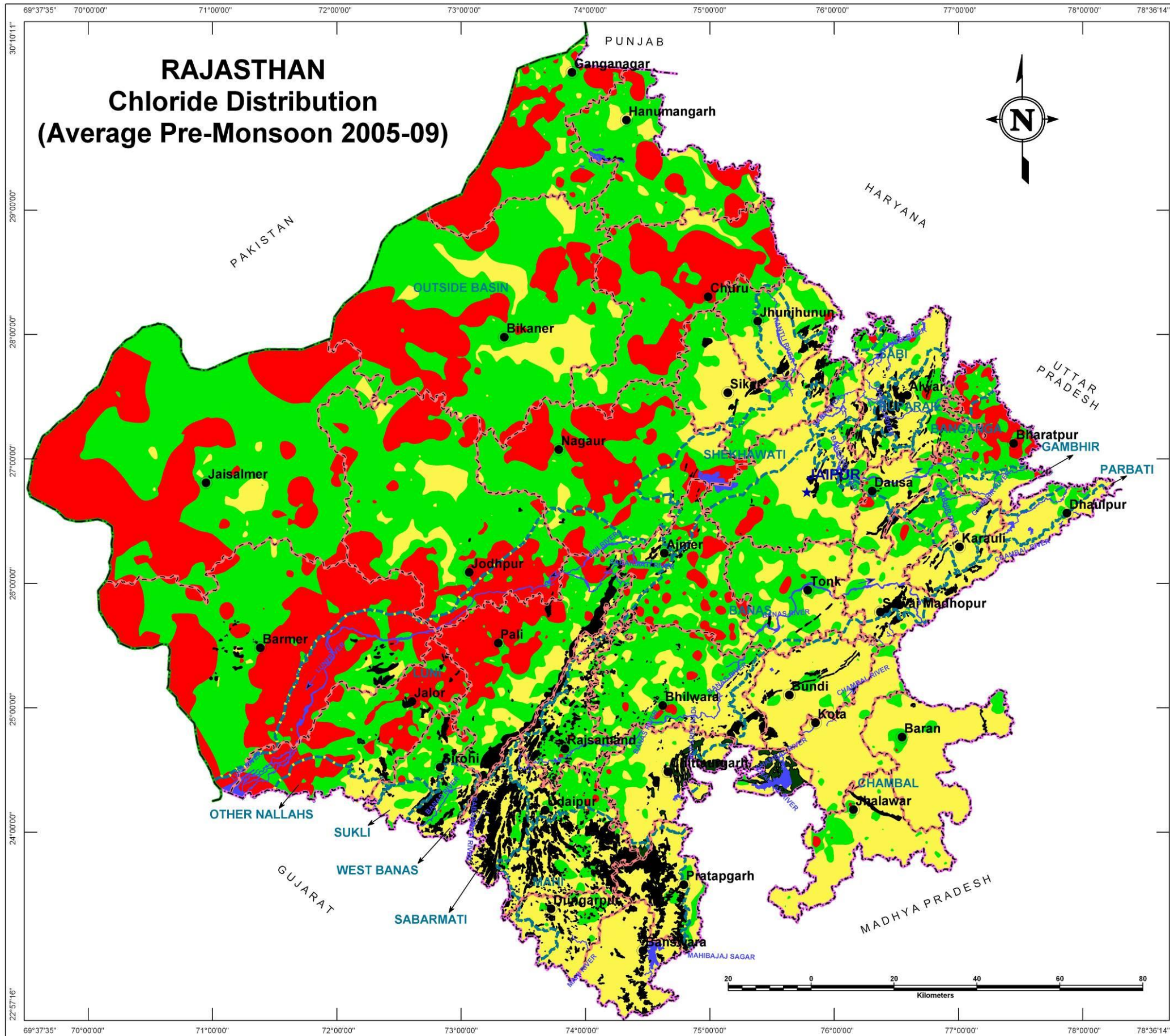
LEGEND

Admin Boundary:

- Capital
- District Headquarter
- International Boundary
- State Boundary
- District Boundary
- Basin Boundary
- Water Bodies:
 - River / Streams
 - Ponds / Reservoirs
- Reserve Forest
- Hills

Chloride Concentration (mg/l):

- < 250
- 250 - 1000
- > 1000



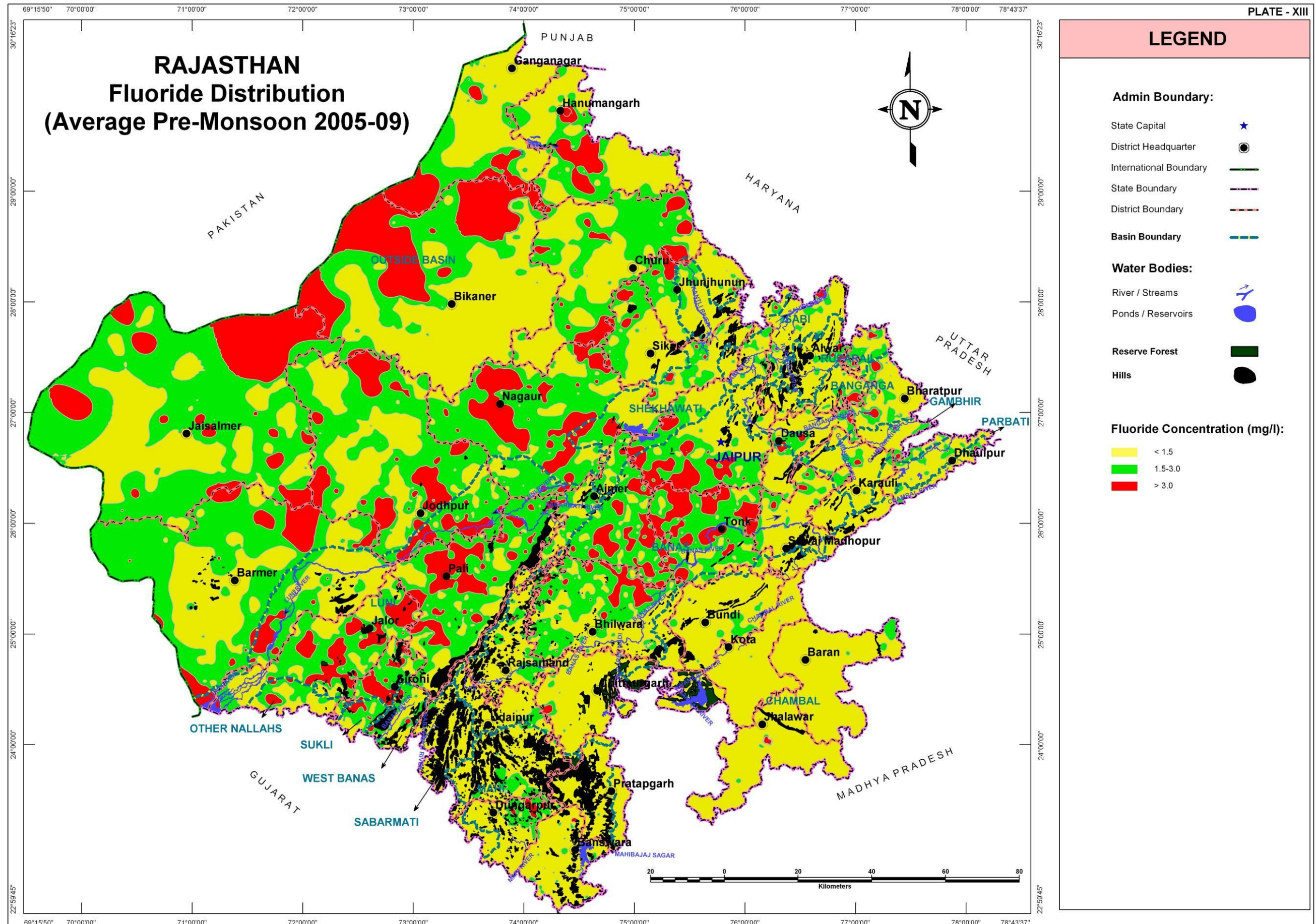
GROUND WATER FLUORIDE DISTRIBUTION

Fluoride distribution map (Plate – XIII) depicts fluoride concentration as observed in ground water samples of wells in the state. High fluoride concentration can be seen on both sides of Aravalli range in Tonk-Alwar-Bhilwara region in the east, whereas, Nagor-Jodhpur-Pali-Jalor belt in the west of the range. In the extreme western part of the state, bordering with Pakistan, also shows very high concentration of fluoride that renders water unusable for human drinking. Apart from specific localized pockets, in general, the areas around Jhalwar-Kota-Baran-Bundi- Chittaurgarh region, Dhaulpur-Karauli-Dausa-Kirauli belt and that around Churu, Bikaner, Barmer and Jaisalmer have by and large have shown very low concentration of fluoride in ground water.

District	Fluoride < 1.5 mg/l		Fluoride 1.5 - 3.0 mg/l		Fluoride > 3.0 mg/l		Total
	Area (sq km)	%age	Area (sq km)	%age	Area (sq km)	%age	
Ajmer	1,734.9	21.1	4,648.5	56.6	1,830.3	22.3	8,213.7
Alwar	5,966.5	80.0	1,142.7	15.3	351.4	4.7	7,460.5
Banswara	3,850.9	94.5	207.3	5.1	15.3	0.4	4,073.5
Baran	6,725.8	96.5	240.4	3.5	-	0.0	6,965.9
Barmer	10,542.0	37.3	14,602.0	51.6	3,135.3	11.1	28,279.3
Bharatpur	3,412.9	67.2	1,405.4	27.7	260.5	5.1	5,078.9
Bhilwara	4,612.7	46.9	4,254.7	43.3	959.6	9.8	9,827.0
Bikaner	14,853.7	49.1	6,780.2	22.4	8,645.3	28.6	30,279.2
Bundi	5,162.1	92.3	428.8	7.7	-	0.0	5,590.9
Chittorgarh	6,148.2	96.3	170.2	2.7	64.8	1.0	6,387.2
Churu	6,437.7	46.5	5,994.9	43.3	1,411.4	10.2	13,844.2
Dausa	2,239.4	69.4	812.8	25.2	174.6	5.4	3,227.0
Dhaulpur	2,393.3	78.7	606.5	19.9	40.6	1.3	3,040.3
Dungarpur	2,618.2	75.0	707.8	20.3	164.6	4.7	3,490.6
Ganganagar	6,134.1	57.4	2,415.6	22.6	2,134.2	20.0	10,684.0
Hanumangarh	5,553.2	56.0	2,818.9	28.4	1,537.5	15.5	9,909.7
Jaipur	5,280.7	49.0	3,558.1	33.0	1,946.6	18.0	10,785.5
Jaisalmer	10,816.1	28.1	20,388.3	53.0	7,282.8	18.9	38,487.2
Jalor	3,408.7	32.7	4,544.0	43.6	2,474.9	23.7	10,427.4
Jhalawar	6,122.2	98.7	49.8	0.8	33.2	0.5	6,205.4
Jhunjhun	3,897.5	69.2	1,406.2	25.0	329.7	5.9	5,633.3
Jodhpur	7,081.1	31.2	11,242.8	49.6	4,362.8	19.2	22,697.0
Karauli	4,258.5	91.2	386.8	8.3	26.4	0.6	4,671.8
Kota	5,105.6	99.7	16.7	0.3	-	0.0	5,122.3
Nagaur	3,768.2	21.2	9,632.9	54.2	4,369.8	24.6	17,770.6
Pali	2,212.6	19.6	6,144.9	54.5	2,907.7	25.8	11,264.8
Pratapgarh	2,900.3	99.3	21.5	0.7	0.3	0.0	2,924.1
Rajsamand	2,913.3	76.9	816.5	21.5	60.2	1.6	3,789.9
Sawai Madhopur	3,553.0	76.0	901.2	19.3	218.0	4.7	4,672.2
Sikar	4,644.9	63.0	2,288.0	31.0	440.7	6.0	7,373.4
Sirohi	1,445.8	33.3	1,941.1	44.8	949.0	21.9	4,335.9
Tonk	1,336.3	18.7	3,535.0	49.4	2,279.8	31.9	7,151.1
Udaipur	6,361.4	87.0	924.4	12.6	23.6	0.3	7,303.5
Total	163,492.0	100.0	115,034.9	100.0	48,430.9	100.0	326,967.3



PLATE - XIII



GROUND WATER NITRATE DISTRIBUTION

The state shows, in general, high concentration of Nitrate in groundwater. The red colour regions represent >100 mg/l of Nitrate in groundwater which is considered unsuitable for agriculture. Major part of western Rajasthan falls under high Nitrate category and good part of Bhilwara-Rajsamand-Udaipur-Dungarpur belt has indicated presence of high Nitrate concentration. In other parts also, pockets of high Nitrate concentration are found widely distributed all over. The areas around Jhalawar-Bundi-Baran belt, around Bikaner, south of Jaisalmer and scattered areas in rest of eastern Rajasthan, low to moderate concentrations of Nitrate are found.

District Name	Nitrate < 50 mg/l		Nitrate 50-100 mg/l		Nitrate > 100 mg/l		Total
	Area (Sq km)	%age	Area (Sq km)	%age	Area (Sq km)	%age	
Ajmer	2,993.6	36.4	3,337.2	40.6	1,882.8	22.9	8,213.7
Alwar	4,425.9	59.3	2,171.1	29.1	863.5	11.6	7,460.5
Banswara	3,049.1	74.9	873.2	21.4	151.3	3.7	4,073.5
Baran	5,374.3	77.1	1,358.3	19.5	233.5	3.4	6,965.9
Barmer	4,975.6	17.6	4,684.9	16.6	18,618.7	65.8	28,279.3
Bharatpur	2,330.7	45.9	1,729.8	34.1	1,018.4	20.1	5,078.9
Bhilwara	3,426.1	34.9	3,275.8	33.3	3,125.0	31.8	9,827.0
Bikaner	8,499.4	28.1	5,330.2	17.6	16,449.7	54.3	30,279.2
Bundi	4,476.0	80.1	968.6	17.3	146.4	2.6	5,590.9
Chittorgarh	2,320.0	36.3	2,444.0	38.3	1,619.2	25.4	6,387.2
Churu	431.4	3.1	1,471.6	10.6	11,941.1	86.3	13,844.2
Dausa	2,074.6	64.3	795.4	24.6	356.9	11.1	3,227.0
Dhaulpur	2,240.3	73.7	558.7	18.4	241.3	7.9	3,040.3
Dungarpur	2,311.4	66.2	783.6	22.4	395.6	11.3	3,490.6
Ganganagar	2,114.3	19.8	3,590.0	33.6	4,979.6	46.6	10,684.0
Hanumangarh	1,419.6	14.3	3,502.2	35.3	4,987.8	50.3	9,909.7
Jaipur	5,681.1	52.7	3,631.9	33.7	1,472.2	13.7	10,785.5
Jaisalmer	15,824.6	41.1	10,417.7	27.1	12,244.9	31.8	38,487.2
Jalor	1,648.2	15.8	3,766.7	36.1	5,012.7	48.1	10,427.4
Jhalawar	3,060.7	49.3	2,822.6	45.5	321.9	5.2	6,205.4
Jhunjhunun	2,044.3	36.3	2,300.6	40.8	1,288.6	22.9	5,633.3
Jodhpur	4,414.8	19.5	8,474.1	37.4	9,797.9	43.2	22,697.0
Karauli	1,371.2	29.4	2,682.9	57.4	617.6	13.2	4,671.8
Kota	3,980.3	77.7	941.7	18.4	200.3	3.9	5,122.3
Nagaur	1,620.3	9.1	4,827.2	27.2	11,323.5	63.7	17,770.6
Pali	4,541.8	40.3	4,267.3	37.9	2,456.1	21.8	11,264.8
Pratapgarh	1,411.9	48.3	1,338.7	45.8	171.5	5.9	2,924.1
Rajsamand	1,433.7	37.8	1,442.5	38.1	913.7	24.1	3,789.9
Sawai Madhopur	1,838.0	39.3	1,768.6	37.9	1,065.6	22.8	4,672.2
Sikar	2,692.3	36.5	2,842.0	38.5	1,839.2	24.9	7,373.4
Sirohi	1,956.2	45.1	1,419.4	32.7	960.4	22.1	4,335.9
Tonk	2,852.7	39.9	2,956.6	41.3	1,341.8	18.8	7,151.1
Udaipur	3,423.7	46.8	1,953.1	26.7	1,932.6	26.4	7,303.5
Total	112,258.3	100.0	94,728.4	100.0	119,971.0	100.0	326,967.3



PLATE - XIV

LEGEND

Admin Boundary:

State Capital



District Headquarter



International Boundary



State Boundary



District Boundary



Basin Boundary



Water Bodies:

River / Streams



Ponds / Reservoirs



Reserve Forest



Hills

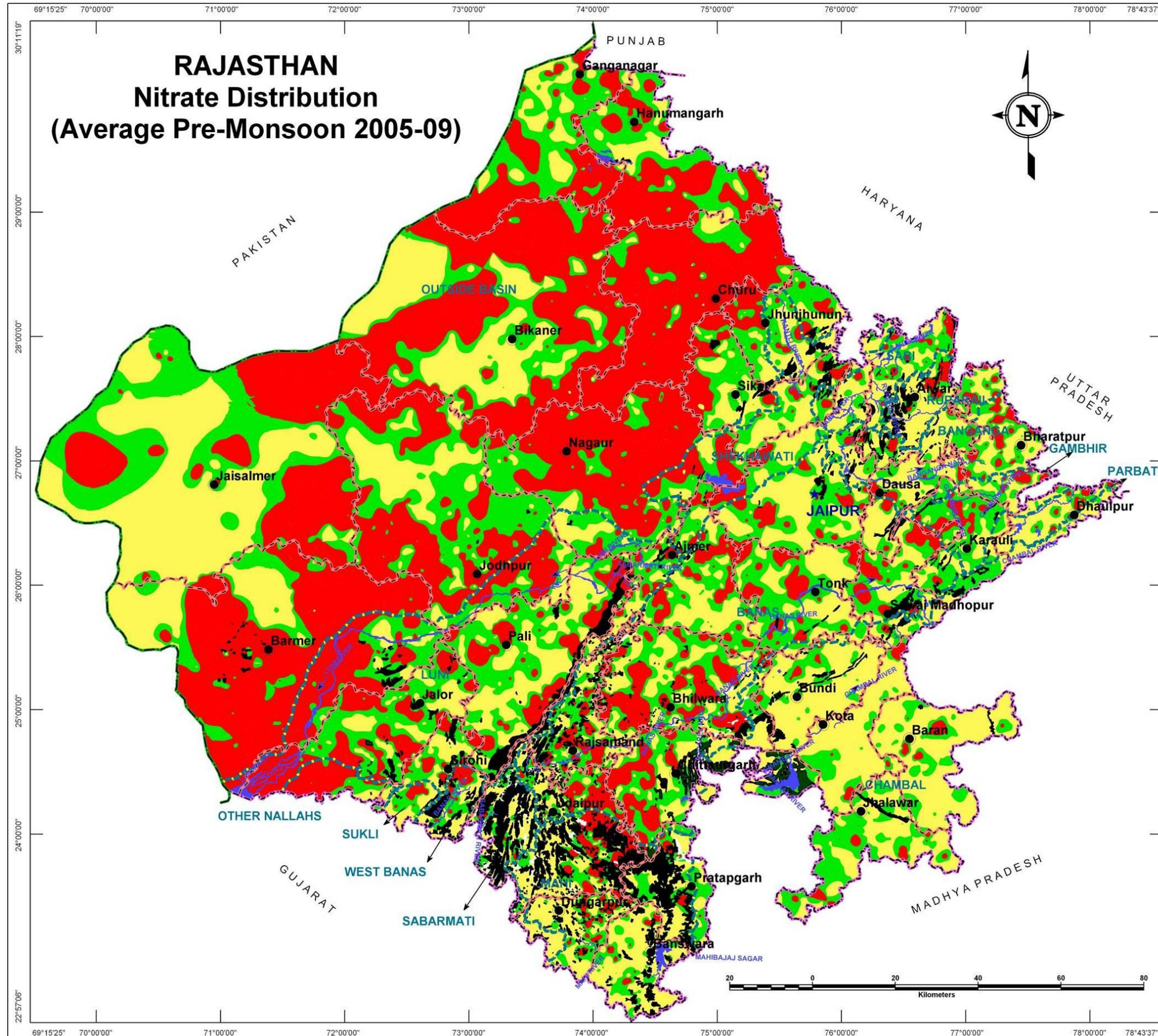


Nitrate Concentration (mg/l):

< 50

50 - 100

> 100



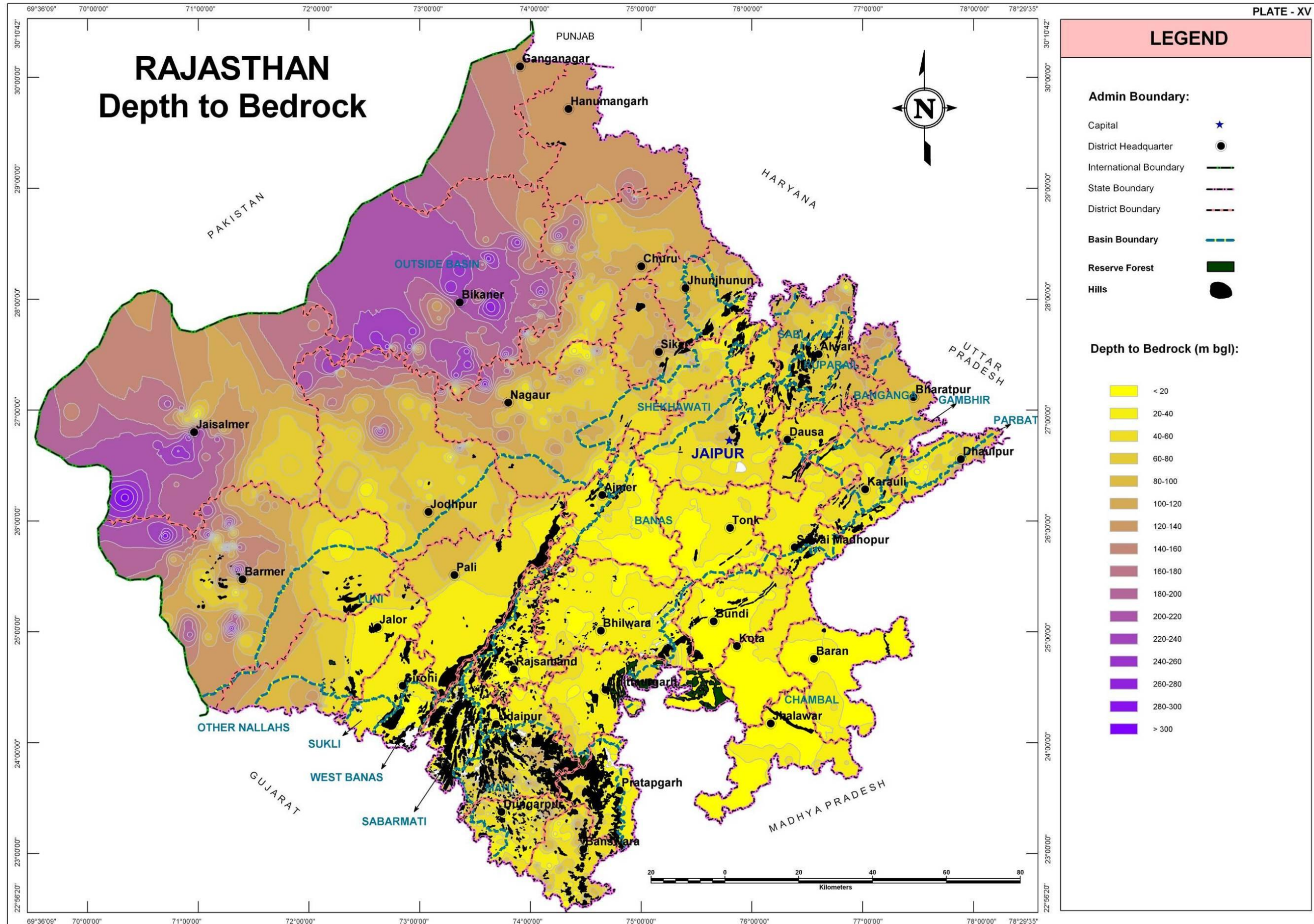
DEPTH TO BEDROCK

From hydrogeological perspective, the beginning of massive bedrock has been considered for defining top of bedrock surface. These rocks are overlain by alluvial deposits of sand, clay, silt and admixture of these in different proportions and thicknesses along with the weathered and fractured zones of the underneath massive bedrock. A perusal of map and data reveals that eastern Rajasthan has shallower bedrock depth whereas in the western part the bedrock occurs at greater depths reaching upto 300m bgl.

District Name	Area (sq km) within depth to bedrock range (m bgl)																Total
	<20 (m)	20-40 (m)	40-60 (m)	60-80 (m)	80-100 (m)	100-120 (m)	120-140 (m)	140-160 (m)	160-180 (m)	180-200 (m)	200-220 (m)	220-240 (m)	240-260 (m)	260-280 (m)	280-300 (m)	>300 (m)	
Ajmer	3,175.5	3,902.1	990.4	145.7	-	-	-	-	-	-	-	-	-	-	-	-	8,213.7
Alwar	-	18.2	1,219.8	3,052.4	2,206.1	744.6	219.4	-	-	-	-	-	-	-	-	-	7,460.5
Banswara	90.5	736.8	1,962.2	964.4	277.3	40.5	1.8	-	-	-	-	-	-	-	-	-	4,073.5
Baran	4,792.1	2,145.3	28.5	-	-	-	-	-	-	-	-	-	-	-	-	-	6,965.9
Barmer	-	624.6	4,478.0	2,206.4	3,178.8	6,283.0	4,491.2	1,931.9	1,220.1	1,815.8	1,367.3	528.3	95.8	42.5	15.6	-	28,279.3
Bharatpur	-	75.3	72.1	1,882.8	2,126.8	689.4	206.0	26.5	-	-	-	-	-	-	-	-	5,078.9
Bhilwara	2,131.5	7,172.6	479.3	40.8	2.8	-	-	-	-	-	-	-	-	-	-	-	9,827.0
Bikaner	-	-	12.9	18.8	35.3	339.1	1,253.1	1,945.1	3,786.2	6,167.7	13,003.2	2,986.1	656.1	75.6	-	-	30,279.2
Bundi	3,356.6	2,070.9	163.4	-	-	-	-	-	-	-	-	-	-	-	-	-	5,590.9
Chittorgarh	1,050.5	3,699.5	1,462.0	169.4	2.8	-	-	-	3.0	-	-	-	-	-	-	-	6,387.2
Churu	0.6	25.8	284.9	1,055.0	3,446.9	5,014.8	2,121.9	1,070.6	579.4	184.1	40.4	13.4	6.4	-	-	-	13,844.2
Dausa	-	191.0	2,260.4	677.3	98.3	-	-	-	-	-	-	-	-	-	-	-	3,227.0
Dhaulpur	33.7	272.5	1,559.3	966.8	136.6	71.4	-	-	-	-	-	-	-	-	-	-	3,040.3
Dungarpur	10.0	1,272.8	1,767.7	314.1	61.8	41.5	22.7	-	-	-	-	-	-	-	-	-	3,490.6
Ganganagar	-	-	-	-	-	-	2,362.1	3,714.9	3,208.5	1,131.4	267.1	-	-	-	-	-	10,684.0
Hanumangarh	-	-	-	-	12.2	305.8	8,996.4	553.5	41.8	-	-	-	-	-	-	-	9,909.7
Jaipur	94.4	2,044.1	5,913.9	2,297.6	435.5	-	-	-	-	-	-	-	-	-	-	-	10,785.5
Jaisalmer	-	19.9	44.2	3,075.2	1,699.8	3,832.8	5,053.3	6,804.2	6,520.0	5,448.3	3,805.8	1,162.2	530.4	284.2	174.4	32.5	38,487.2
Jalor	-	2,872.1	1,485.2	1,678.5	2,124.8	1,358.2	908.6	-	-	-	-	-	-	-	-	-	10,427.4
Jhalawar	4,352.3	1,756.2	96.9	-	-	-	-	-	-	-	-	-	-	-	-	-	6,205.4
Jhunjhun	-	110.7	452.5	733.5	2,834.2	1,479.4	23.0	-	-	-	-	-	-	-	-	-	5,633.3
Jodhpur	10.4	152.0	1,398.0	7,658.8	6,367.3	1,509.9	1,375.9	1,422.3	1,086.6	928.4	566.8	220.6	-	-	-	-	22,697.0
Karauli	70.3	1,257.0	2,350.3	935.7	58.5	-	-	-	-	-	-	-	-	-	-	-	4,671.8
Kota	3,290.0	1,832.3	-	-	-	-	-	-	-	-	-	-	-	-	-	-	5,122.3
Nagaur	26.4	393.4	1,454.2	3,980.1	6,312.3	2,805.4	1,639.4	527.8	298.9	173.7	92.2	35.4	23.3	8.1	-	-	17,770.6
Pali	26.0	5,313.6	2,826.4	1,727.5	1,371.3	-	-	-	-	-	-	-	-	-	-	-	11,264.8
Pratapgarh	283.3	1,651.4	799.0	182.8	7.6	-	-	-	-	-	-	-	-	-	-	-	2,924.1
Rajsamand	909.4	2,513.9	336.3	16.5	13.8	-	-	-	-	-	-	-	-	-	-	-	3,789.9
Sawai Madhopur	2.6	3,520.3	1,102.4	46.9	-	-	-	-	-	-	-	-	-	-	-	-	4,672.2
Sikar	-	620.2	1,223.9	1,658.2	1,745.8	1,885.6	239.7	-	-	-	-	-	-	-	-	-	7,373.4
Sirohi	634.4	3,546.1	155.4	-	-	-	-	-	-	-	-	-	-	-	-	-	4,335.9
Tonk	2,179.2	4,860.5	111.4	-	-	-	-	-	-	-	-	-	-	-	-	-	7,151.1
Udaipur	885.5	2,891.6	1,723.7	1,114.6	496.4	191.7	-	-	-	-	-	-	-	-	-	-	7,303.5
Total	27,405.2	57,562.7	38,214.6	36,599.8	35,053.0	26,593.1	28,914.5	17,996.8	16,744.5	15,849.4	19,142.8	4,946.0	1,312.0	410.4	190.0	32.5	326,967.3



PLATE - XV



UNCONFINED AQUIFER

Unconfined aquifer in alluvial areas

Alluvial aquifers occur mostly in north, northeast and southwestern parts of the state. These attain significant thickness in Hanumangarh, Jalor, Bikaner and Barmer districts whereas spatially they are most prominent aquifers in Bikaner, Barmer, Jaisalmer, Ganganagar, Hanumangarh, Jalor and Churu where their total area is around 9,000 sq kms or more reaching upto 15,400 sq kms (in Bikaner).

Aquifer Thickness (m)	Area (sq km) within aquifer thickness range (m)											Total
	< 10	10-20	20-30	30-40	40-50	50-60	60-70	70-80	80-90	90-100	>100	
Ajmer	392.4	0.1	-	-	-	-	-	-	-	-	-	392.5
Alwar	3,031.1	1,953.9	652.7	258.5	57.2	6.5	1.1	0.5	-	-	-	5,961.5
Banswara	-	-	-	-	-	-	-	-	-	-	-	-
Baran	326.9	-	-	-	-	-	-	-	-	-	-	326.9
Barmer	8,284.9	3,097.8	1,646.0	1,527.3	441.1	159.7	82.1	49.7	32.6	22.6	6.2	15,350.0
Bharatpur	1,369.8	1,626.8	1,020.4	389.0	200.1	109.9	44.4	3.3	-	-	-	4,763.7
Bhilwara	-	-	-	-	-	-	-	-	-	-	-	-
Bikaner	1,714.7	2,836.2	5,042.1	2,765.9	1,268.6	888.0	600.9	281.5	6.0	-	-	15,403.9
Bundi	747.4	516.4	-	-	-	-	-	-	-	-	-	1,263.8
Chittorgarh	-	-	-	-	-	-	-	-	-	-	-	-
Churu	2,259.0	2,910.9	1,881.4	1,200.0	631.9	358.3	142.0	59.4	24.4	6.4	0.1	9,473.8
Dausa	2,156.7	471.4	90.3	17.4	-	-	-	-	-	-	-	2,735.8
Dhaulpur	1,103.9	427.3	68.4	-	-	-	-	-	-	-	-	1,599.6
Dungarpur	-	-	-	-	-	-	-	-	-	-	-	-
Ganganagar	442.5	2,634.2	2,224.0	1,946.7	1,319.0	1,200.4	713.5	173.5	30.2	-	-	10,684.0
Hanumangarh	402.2	1,422.2	1,801.3	3,636.9	1,716.8	453.4	201.0	117.2	77.2	42.7	38.8	9,909.7
Jaipur	5,578.8	1,388.6	512.6	76.6	3.4	-	-	-	-	-	-	7,560.0
Jaisalmer	3,398.5	1,679.2	2,460.5	216.1	1,661.5	1,366.2	1,375.8	1,883.0	113.9	7.3	-	14,162.0
Jalor	4,542.7	1,287.5	918.9	835.2	999.9	683.5	240.7	42.8	29.4	24.9	45.9	9,651.4
Jhalawar	-	-	-	-	-	-	-	-	-	-	-	-
Jhunjhun	3,114.9	1,006.0	436.2	217.6	54.0	1.6	-	-	-	-	-	4,830.3
Jodhpur	2,450.2	951.7	95.6	13.3	-	-	-	-	-	-	-	3,510.8
Karauli	1,522.0	463.7	311.4	69.7	8.7	-	-	-	-	-	-	2,375.5
Kota	1,342.4	207.0	-	-	-	-	-	-	-	-	-	1,549.4
Nagaur	3,823.7	1,553.5	713.6	292.8	100.1	38.7	19.9	16.6	3.3	-	-	6,562.2
Pali	2,006.8	830.1	304.6	98.0	48.0	25.7	14.8	7.8	3.6	0.8	-	3,340.2
Pratapgarh	-	-	-	-	-	-	-	-	-	-	-	-
Rajsamand	-	-	-	-	-	-	-	-	-	-	-	-
Sawai Madhopur	1,567.6	711.5	289.4	60.7	2.1	-	-	-	-	-	-	2,631.3
Sikar	1,516.1	1,821.3	1,370.5	715.8	430.9	161.9	19.1	9.2	-	-	-	6,044.8
Sirohi	847.2	103.8	54.1	19.9	-	-	-	-	-	-	-	1,025.0
Tonk	1,336.5	87.7	4.7	-	-	-	-	-	-	-	-	1,428.9
Udaipur	-	-	-	-	-	-	-	-	-	-	-	-
Total	55,278.9	29,988.8	21,898.7	14,357.4	8,943.3	5,453.8	3,455.3	2,644.5	320.6	104.7	91.0	1,42,537.0

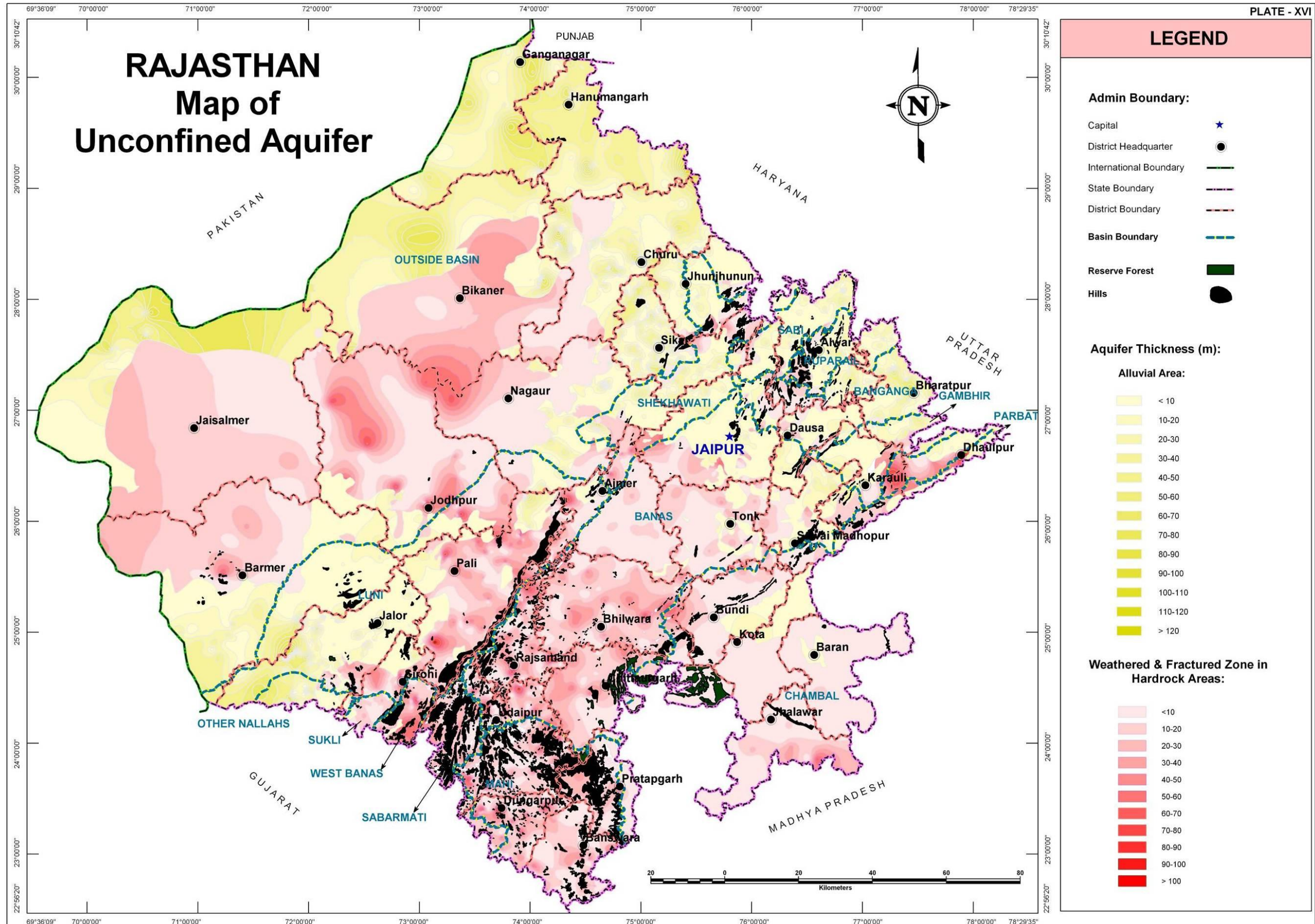
Unconfined aquifer hardrock areas

The weathered and fractured hardrocks (both consolidated and semi-consolidated) also constitute good aquifers in Central, southeast and western parts of the state. While Pali (90-100m thickness), Dhaulpur, Rajsamand seem to have significant thickness of such zones, they are significant in terms of spatial coverage in Jaisalmer, Jodhpur, Bikaner, Barmer, Nagaur, Bhilwara etc. having more than 9,000 sq kms area and reaching upto 24,300 sq kms (in Jaisalmer). It is also important to note that Bhilwara, Udaipur, Chittaurgarh, Jhalwar, Banswara, Rajsamand, Dungarpur etc. do not have any significant alluvial aquifers and ground water is mostly drawn from hardrock aquifers only.

Aquifer Thickness (m)	Area (sq km) within aquifer thickness range (m)										Total
	<10	10-20	20-30	30-40	40-50	50-60	60-70	70-80	80-90	90-100	
Ajmer	5,205.3	1,755.5	467.8	185.6	98.9	55.2	29.8	18.5	4.6	-	7,821.2
Alwar	850.8	638.3	9.9	-	-	-	-	-	-	-	1,499.0
Banswara	1,642.7	1,585.9	550.5	252.3	28.1	14.0	-	-	-	-	4,073.5
Baran	6,498.9	140.1	-	-	-	-	-	-	-	-	6,639.0
Barmer	3,085.3	6,472.5	3,039.6	242.0	62.9	23.3	3.7	-	-	-	12,929.3
Bharatpur	142.1	172.9	0.2	-	-	-	-	-	-	-	315.2
Bhilwara	1,832.7	4,213.4	2,327.6	1,021.1	312.1	93.7	25.3	1.1	-	-	9,827.0
Bikaner	1,059.1	5,658.6	4,470.2	2,770.8	612.1	304.5	-	-	-	-	14,875.3
Bundi	3,114.2	827.5	264.8	98.6	21.2	0.8	-	-	-	-	4,327.1
Chittorgarh	2,122.8	2,051.5	1,404.4	719.9	88.6	-	-	-	-	-	6,387.2
Churu	3,512.8	758.4	99.2	-	-	-	-	-	-	-	4,370.4
Dausa	329.2	114.9	39.0	8.1	-	-	-	-	-	-	491.2
Dhaulpur	184.7	197.8	168.6	265.1	286.7	143.6	89.9	61.7	35.5	7.1	1,440.7
Dungarpur	661.7	1,222.2	999.6	363.8	137.0	62.3	28.5	11.4	4.1	-	3,490.6
Ganganagar	-	-	-	-	-	-	-	-	-	-	-
Hanumangarh	-	-	-	-	-	-	-	-	-	-	-
Jaipur	1,256.9	1,043.6	482.0	285.2	113.6	44.2	-	-	-	-	3,225.5
Jaisalmer	6,965.6	12,695.7	4,663.9	-	-	-	-	-	-	-	24,325.2
Jalor	185.3	253.6	259.3	74.5	3.3	-	-	-	-	-	776.0
Jhalawar	4,031.4	1,126.8	683.0	245.3	72.5	44.1	2.3	-	-	-	6,205.4
Jhunjhunun	511.4	148.0	54.9	60.5	23.3	4.0	0.9	-	-	-	803.0
Jodhpur	7,491.1	4,501.9	3,003.3	1,885.0	1,515.1	681.5	101.8	6.5	-	-	19,186.2
Karauli	403.2	1,135.4	215.4	210.7	142.2	98.9	51.5	27.2	11.5	0.3	2,296.3
Kota	3,570.7	2.2	-	-	-	-	-	-	-	-	3,572.9
Nagaur	6,697.9	2,719.1	848.7	403.6	244.3	263.5	31.3	-	-	-	11,208.4
Pali	1,169.5	2,607.1	2,387.6	1,197.7	357.8	103.3	49.4	25.8	15.0	11.4	7,924.6
Pratapgarh	1,619.6	705.2	495.4	101.8	2.1	-	-	-	-	-	2,924.1
Rajsamand	790.8	1,114.6	1,021.9	529.6	200.9	69.6	31.2	17.9	10.2	3.2	3,789.9
Sawai Madhopur	1,320.5	582.0	121.5	16.9	-	-	-	-	-	-	2,040.9
Sikar	712.0	479.1	102.8	26.7	5.8	2.2	0.0	-	-	-	1,328.6
Sirohi	662.5	1,217.3	507.2	423.3	269.3	209.4	15.4	3.7	2.2	0.6	3,310.9
Tonk	3,322.0	1,998.7	345.1	45.3	10.3	0.8	-	-	-	-	5,722.2
Udaipur	1,598.7	2,736.3	1,732.1	740.9	299.5	134.7	37.7	15.0	5.3	3.3	7,303.5
Total	72,551.4	60,876.1	30,765.5	12,174.2	4,907.6	2,353.6	498.7	188.8	88.4	25.9	1,84,430.3



PLATE - XVI



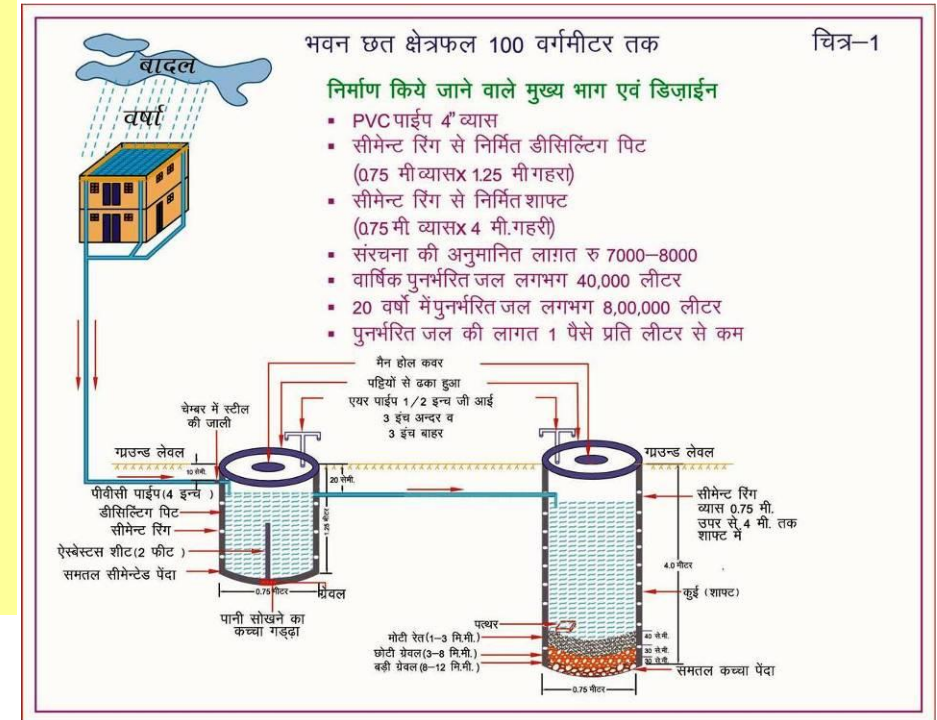
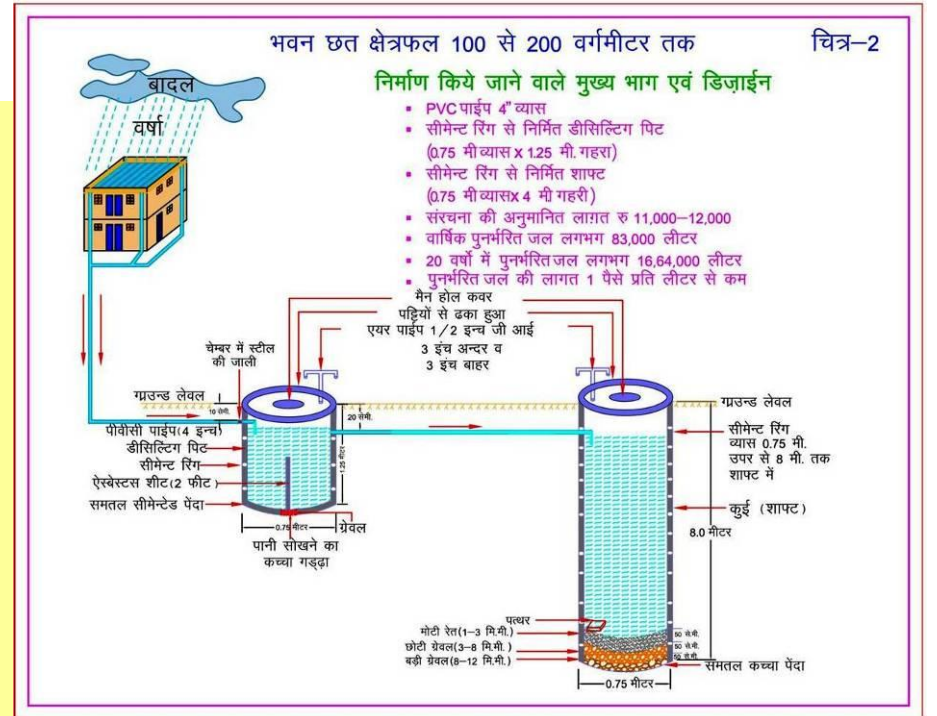
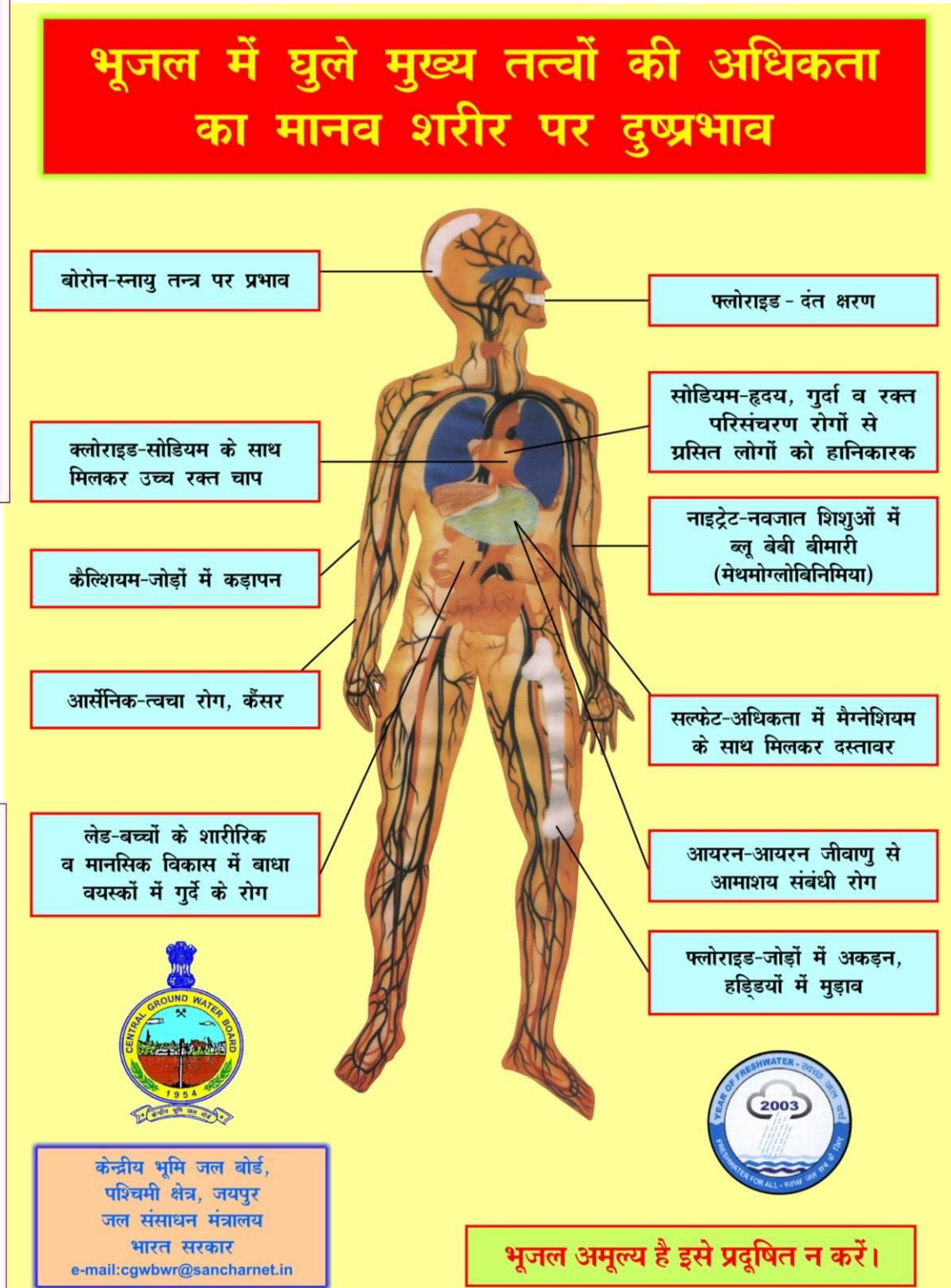
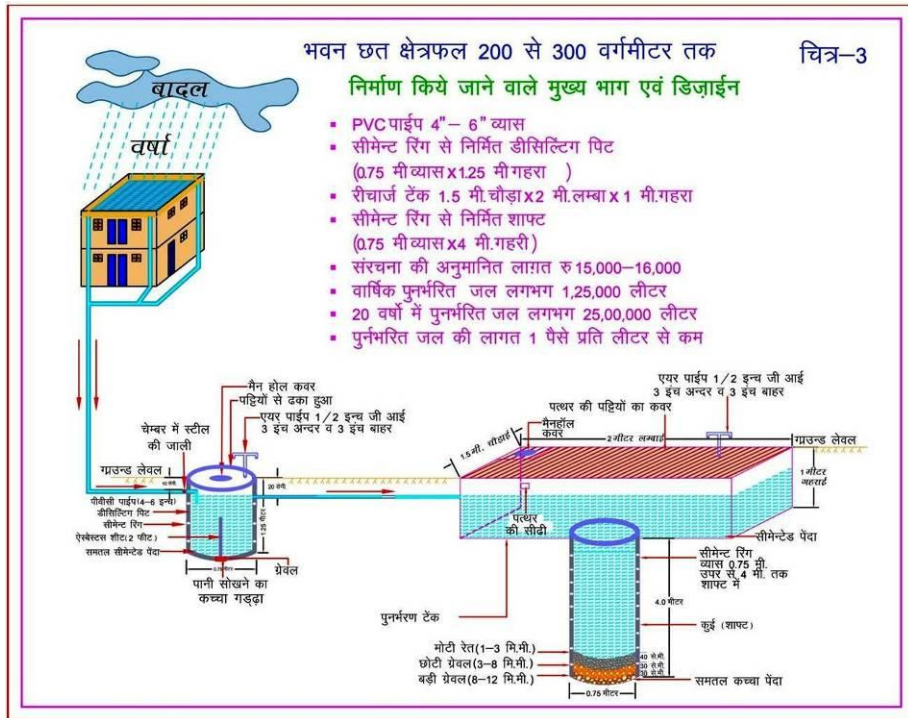
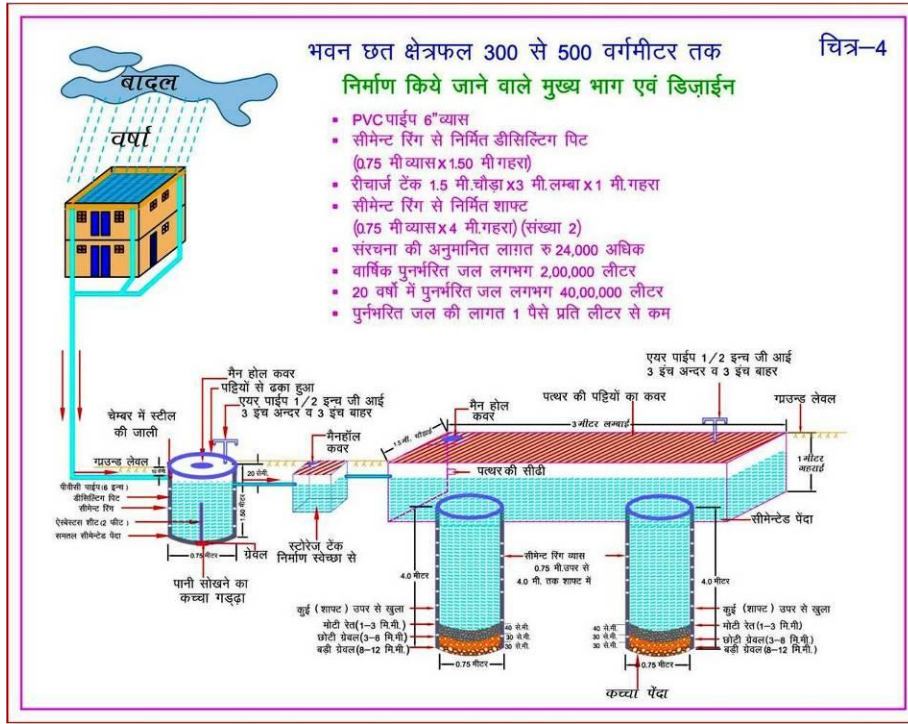


Glossary of terms

S. No.	Technical Terms	Definition
1	AQUIFER	A saturated geological formation which has good permeability to supply sufficient quantity of water to a Tube well, well or spring.
2	ARID CLIMATE	Climate characterized by high evaporation and low precipitation.
3	ARTIFICIAL RECHARGE	Addition of water to a groundwater reservoir by man made activity
4	CLIMATE	The sum total of all atmospheric or meteorological influences principally temperature, moisture, wind, pressure and evaporation of a region.
5	CONFINED AQUIFER	A water bearing strata having confined impermeable overburden. In this aquifer, water level represent the piezometric head.
6	CONTAMINATION	Introduction of undesirable substance, normally not found in water, which renders the water unfit for its intended use.
7	DRAWDOWN	The drawdown is the depth by which water level is lowered.
8	FRESH WATER	Water suitable for drinking purpose.
9	GROUND WATER	Water found below the land surface.
10	GROUND WATER BASIN	A hydro-geologic unit containing one large aquifer or several connected and interrelated aquifers.
11	GROUNDWATER RECHARGE	The natural infiltration of surface water into the ground.
12	HARD WATER	The water which does not produce sufficient foam with soap.
13	HYDRAULIC CONDUCTIVITY	A constant that serves as a measure of permeability of porous medium.
14	HYDROGEOLOGY	The science related with the ground water.
15	HUMID CLIMATE	The area having high moisture content.
16	ISOHYET	A line of equal amount of rainfall.
17	METEOROLOGY	Science of the atmosphere.
18	PERCOLATION	It is flow through a porous substance.
19	PERMEABILITY	The property or capacity of a soil or rock for transmitting water.
20	pH	Value of hydrogen-ion concentration in water. Used as an indicator of acidity (pH < 7) or alkalinity (pH > 7).
21	PIEZOMETRIC HEAD	Elevation to which water will rise in a piezometers.
22	RECHARGE	It is a natural or artificial process by which water is added from outside to the aquifer.
23	SAFE YIELD	Amount of water which can be extracted from groundwater without producing undesirable effect.
24	SALINITY	Concentration of dissolved salts.
25	SEMI-ARID	An area is considered semiarid having annual rainfall between 10-20 inches.
26	SEMI-CONFINED AQUIFER	Aquifer overlain and/or underlain by a relatively thin semi-pervious layer.
27	SPECIFIC YIELD	Quantity of water which is released by a formation after it's complete saturation.
28	TOTAL DISSOLVED SOLIDS	Total weight of dissolved mineral constituents in water per unit volume (or weight) of water in the sample.

(Contd...)

S. No.	Technical Terms	Definition
29	TRANSMISSIBILITY	It is defined as the rate of flow through an aquifer of unit width and total saturation depth under unit hydraulic gradient. It is equal to product of full saturation depth of aquifer and its coefficient of permeability.
30	UNCONFINED AQUIFER	A water bearing formation having permeable overburden. The water table forms the upper boundary of the aquifer.
31	UNSATURATED ZONE	The zone below the land surface in which pore space contains both water and air.
32	WATER CONSERVATION	Optimal use and proper storage of water.
33	WATER RESOURCES	Availability of surface and ground water.
34	WATER RESOURCES MANAGEMENT	Planned development, distribution and use of water resources.
35	WATER TABLE	Water table is the upper surface of the zone of saturation at atmospheric pressure.
36	ZONE OF SATURATION	The ground in which all pores are completely filled with water.
37	ELECTRICAL CONDUCTIVITY	Flow of free ions in the water at 25C mu/cm.
38	CROSS SECTION	A Vertical Projection showing sub-surface formations encountered in a specific plane.
39	3-D PICTURE	A structure showing all three dimensions i.e. length, width and depth.
40	GWD	Ground Water Department
41	CGWB	Central Ground Water Board
42	CGWA	Central Ground Water Authority
43	SWRPD	State Water Resources Planning Department
44	EU-SPP	European Union State Partnership Programme
45	TOPOGRAPHY	Details of drainage lines and physical features of land surface on a map.
46	GEOLOGY	The science related with the Earth.
47	GEOMORPHOLOGY	The description and interpretation of land forms.
48	PRE MONSOON SURVEY	Monitoring of Ground Water level from the selected DKW/Piezometer before Monsoon (carried out between 15th May to 15th June)
49	POST-MONSOON SURVEY	Monitoring of Ground Water level from the selected DKW/Piezometer after Monsoon (carried out between 15th October to 15th November)
50	PIEZOMETER	A non-pumping small diameter bore hole used for monitoring of static water level.
51	GROUND WATER FLUCTUATION	Change in static water level below ground level.
52	WATER TABLE	The static water level found in unconfined aquifer.
53	DEPTH OF BED ROCK	Hard & compact rock encountered below land Surface.
54	G.W. MONITORING STATION	Dug wells selected on grid basis for monitoring of state water level.
55	EOLIAN DEPOSITS	Wind blown sand deposits



Myths and Facts about Ground Water

S No	Myths	Facts
1	What is Ground Water • an underground lake • a net work of underground rivers • a bowl filled with water	Water which occurs below the land in geological formations/rocks is Ground water
2	Ground Water occurs everywhere beneath the Land Surface	Not really, it depends on the nature of rock formation
3	There is a relationship between ground water and surface water	Not all the places. Near streams/rivers there is relation
4	Groundwater is not renewable resource	It is renewable source and every year it is being recharged through rain/applied irrigation etc
5	Ground water is unlimited and deeper you drill more discharge	It is limited to annual recharge from rain/applied irrigation. The discharge may not increase if you go deeper
6	Ground Water moves rapidly	The movement of ground water is very slow
7	Ground water pumped from wells is thousands of years old	Generally the ground water being tapped through wells is a few years old
8	If water taste good—it is safe to drink	It may have other chemicals e.g. fluoride, nitrates etc which are harmful
9	Water from free flowing tube wells is very pure	This water can also be contaminated so test before use
10	If I recharge my TW/DW/HP it will not benefit me	It will also benefit you and also adjoining wells
11	There is no static ground water resources in Rajasthan	Rajasthan is also having Static GW resources, and being tapped in most of areas as GW annual withdrawal is more than annual recharge
12	I cannot meet annual cooking and drinking water requirement by rain water harvesting	The water requirement for drinking and cooking is only 8 lit/day. You can harvest this water for family of 5 persons from roof top or paved area of 75 Sq m to meet annual requirement
13	You can increase ground water recharge	This can be done by harvesting the rain water and storing in sub surface reservoir (GW) by constructing the recharge structures
14	You cannot use abandoned TW/HP/DW for ground water recharge	These should be used as recharge structures as harvested rain water is directly put into GW reservoir
15	Putting waste near HP/TW will not cause any problem	Such actions will pollute wells and water



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